Dynamics of the spike-specific B-cell repertoire following SARS-CoV-2 Vaccination

Thomas Dupic

April 10th 2026, IRCM

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The B-cell Repertoire

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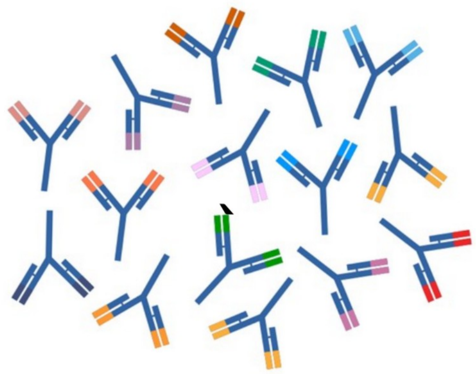
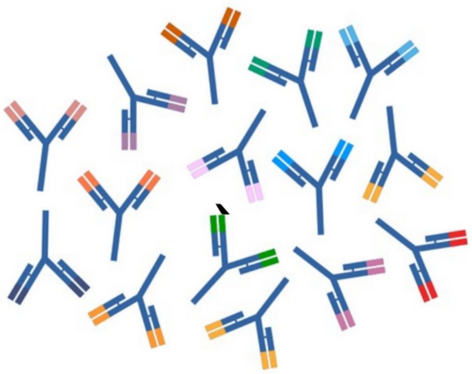


Set of all **B-cell receptors (BCRs)**.

The B-cell Repertoire



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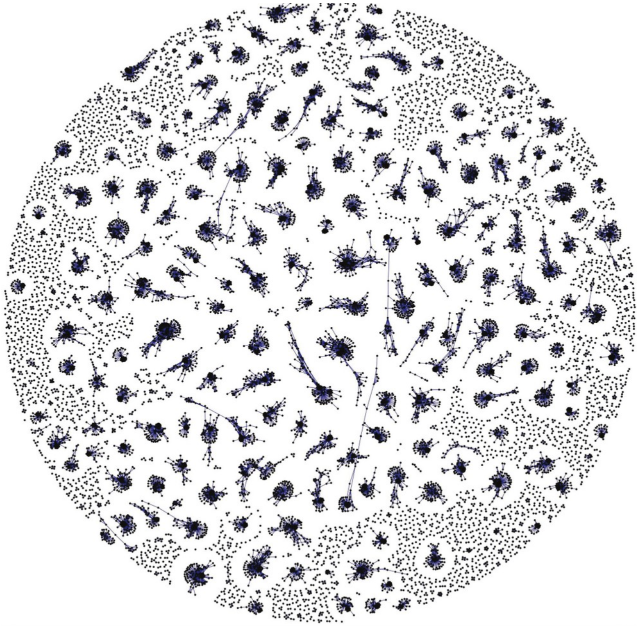
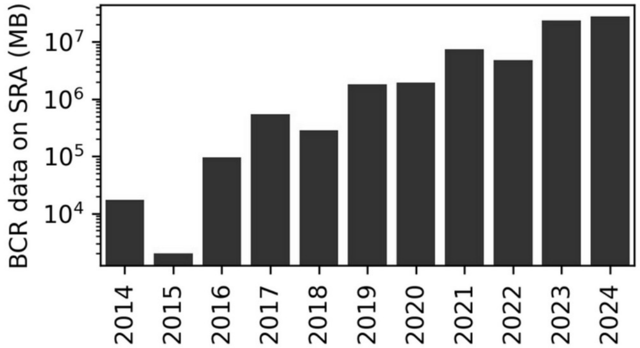


Extremely high diversity, at least 10^9 different receptors in one individual.

The B-cell Repertoire



BCR genes are easily sequenced from bulk RNA (RepSeq)

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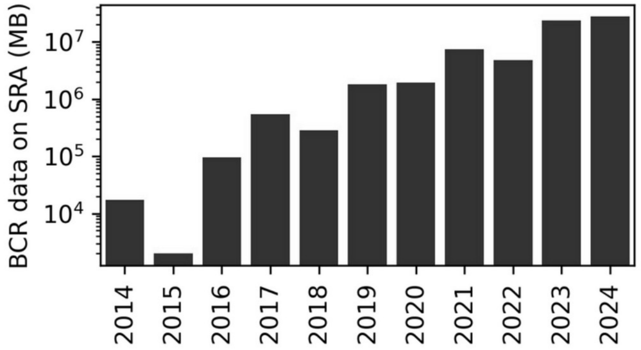

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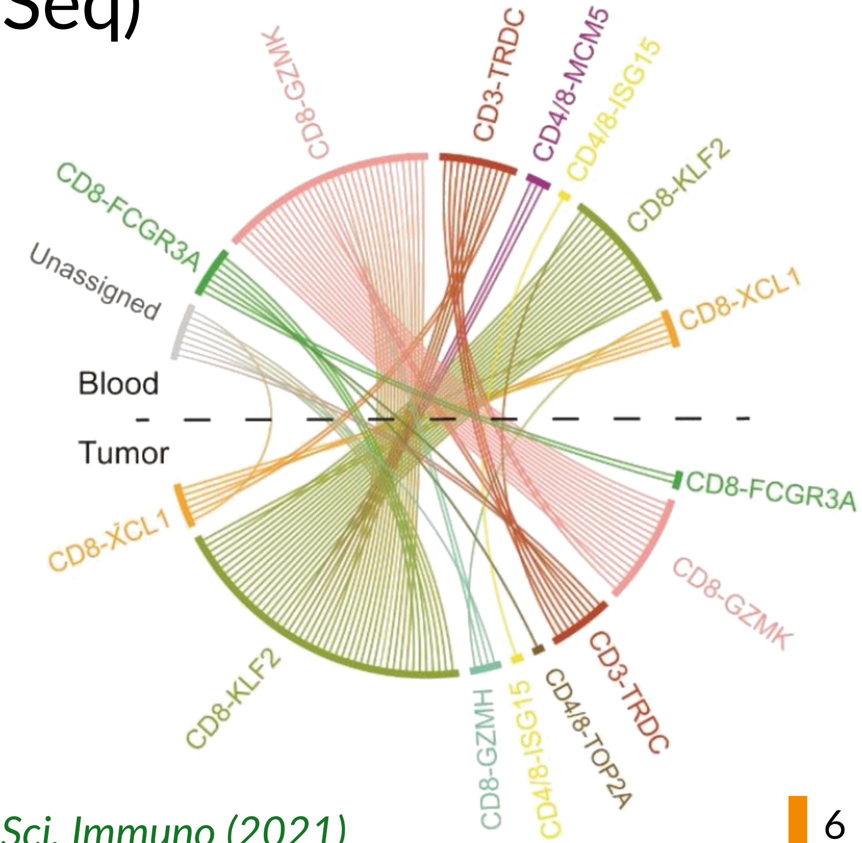
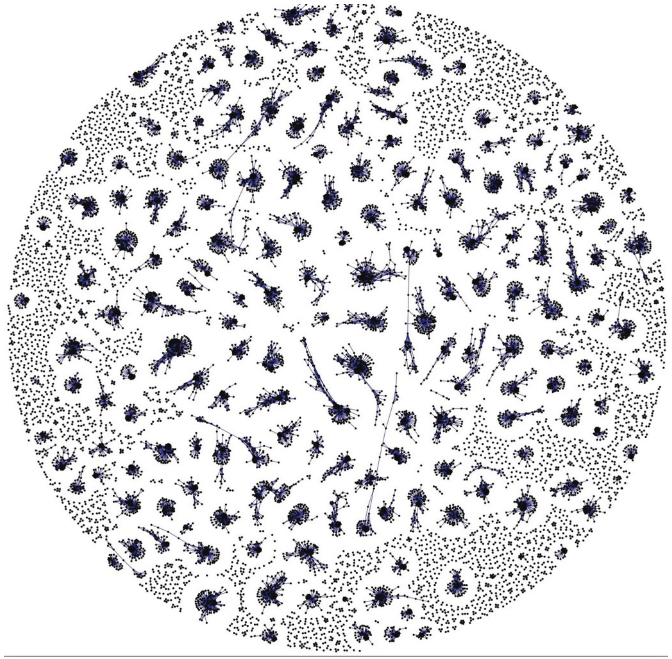
Also common in single-cell sequencing: “Clonal barcode”



Gueguen, Metoikidou, Dupic et al, Sci. Immuno (2021)

The Potential of the Repertoire

⇒ Prognosis



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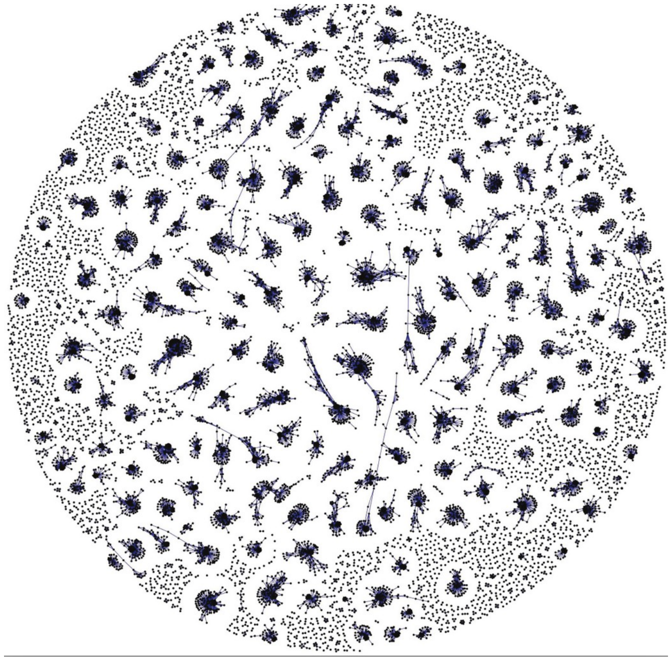

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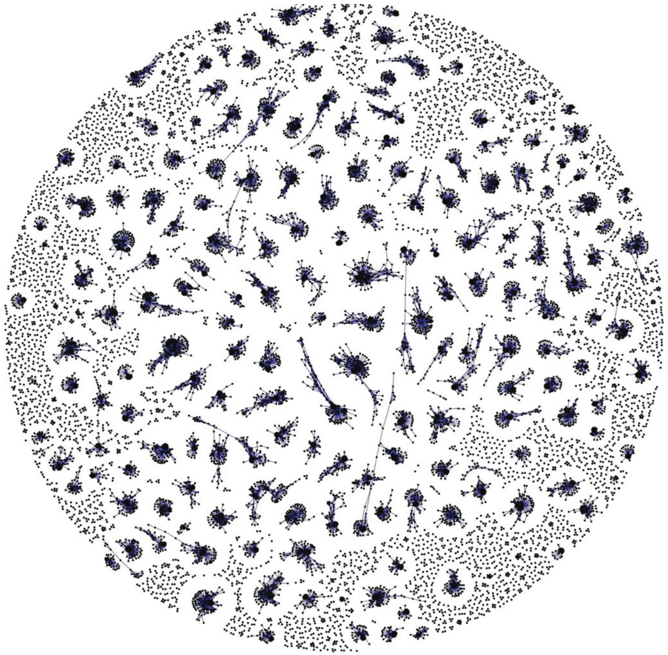

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⇒ Therapeutic Antibodies

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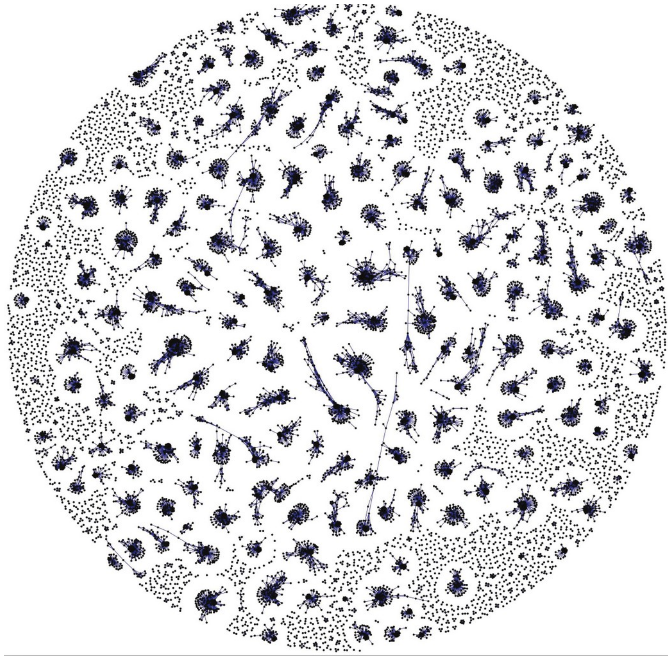

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Marta Łuksza, Zachary M. Sethna, Luis A. Rojas, Jayon Lihm, Barbara Bravi, Yuval Elhanati, Kevin Soares, Masataka Amisaki, Anton Dobrin, David Hoyos, Pablo Guasp, Abderezak Zebboudj, Rebecca Yu, Adrienne Kaya Chandra, Theresa Waters, Zagaa Odgerel, Joanne Leung, Rajya Kappagantula, Alvin Makohon-Moore, Amber Johns, Anthony Gill, Mathieu Gigoux, Jedd Wolchok, Taha Merghoub, … Vinod P. Balachandran

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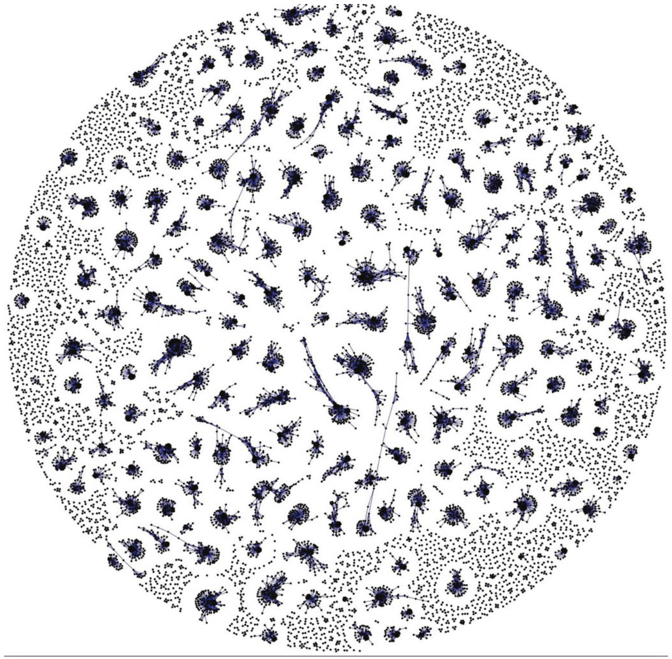

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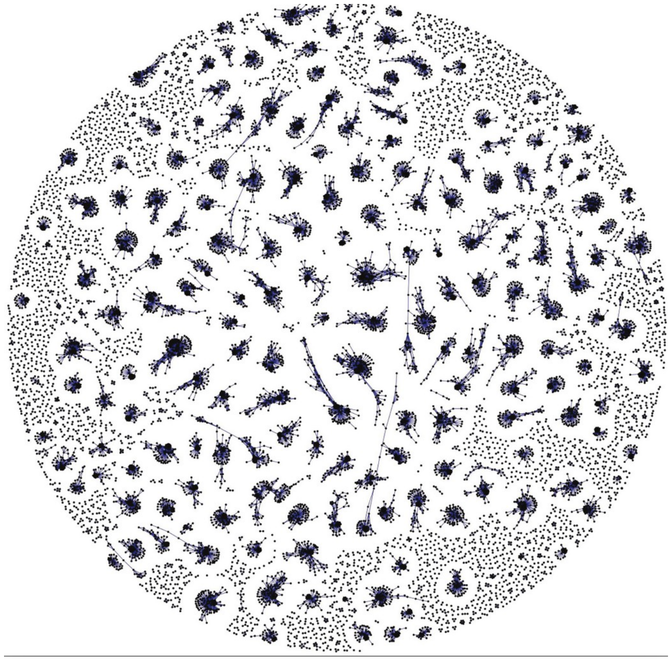

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Functional HIV-1/HCV cross-reactive antibodies isolated from a chronically co-infected donor

Kelsey A. Pilewski [1,2,17] · Steven Wall [1,2] · Simone I. Richardson [3,4] · … · Spyros Kalams [2] · Lynn Morris [3,4] · Ivelin S. Georgiev [1,2,13,14,15,16,19] … Show more

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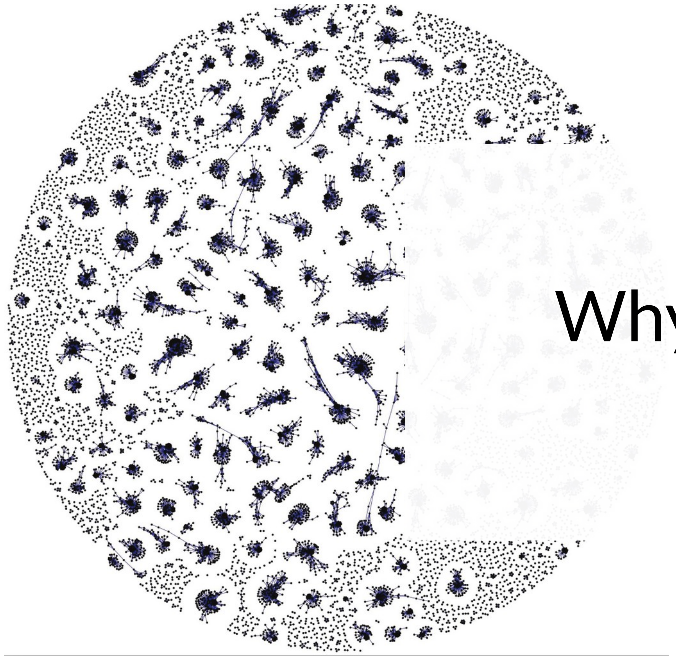

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⇒ Diagnostic & Medical history

Why are we not there yet ?

Disease diagnostics using machine learning of B cell and T cell receptor sequences

…, JI-YEUN LEE, KHOA D. NGUYEN, RAMONA A. HOH

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Repertoires are complicated to analyze

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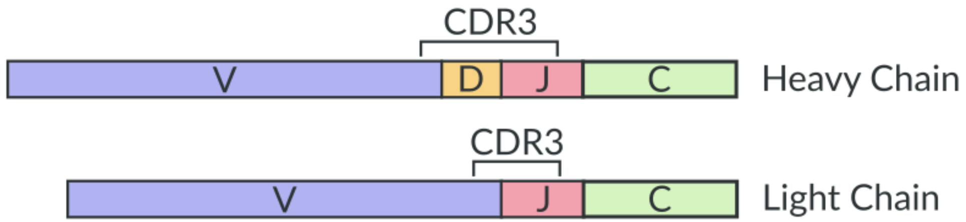
BCRs are randomly generated by V(D)J recombination.



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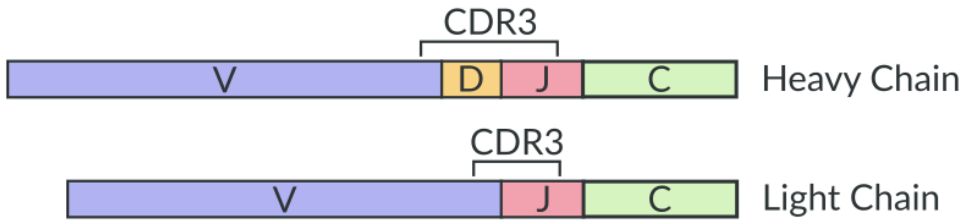

→ Each individual repertoire is unique
→ Most BCRs are not shared
→ Not all BCRs are as likely

Statistical Problems

B-cell Repertoires are **particularly** complicated to study

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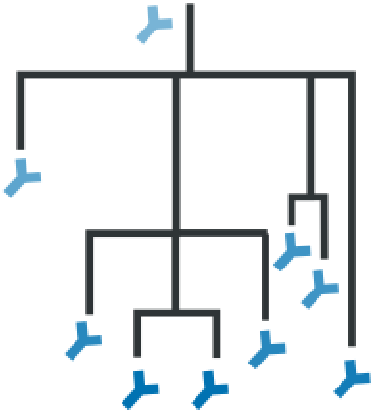

BCRs can evolve against antigens and acquire mutations

Statistical Problems

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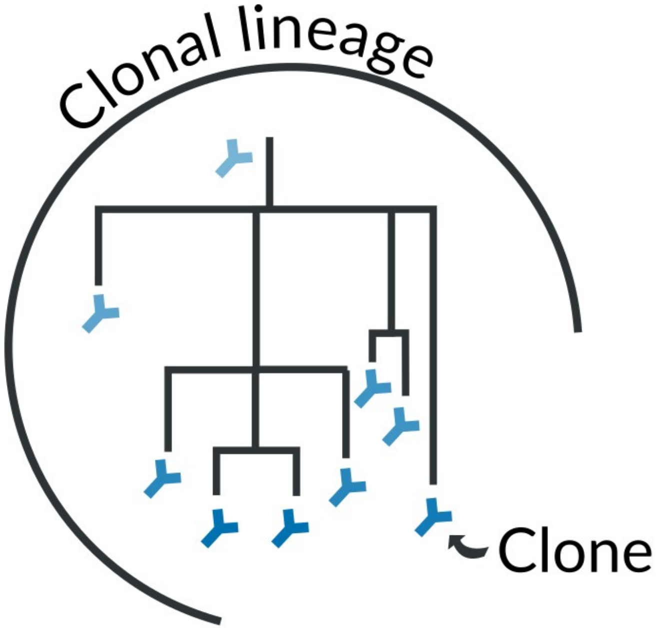

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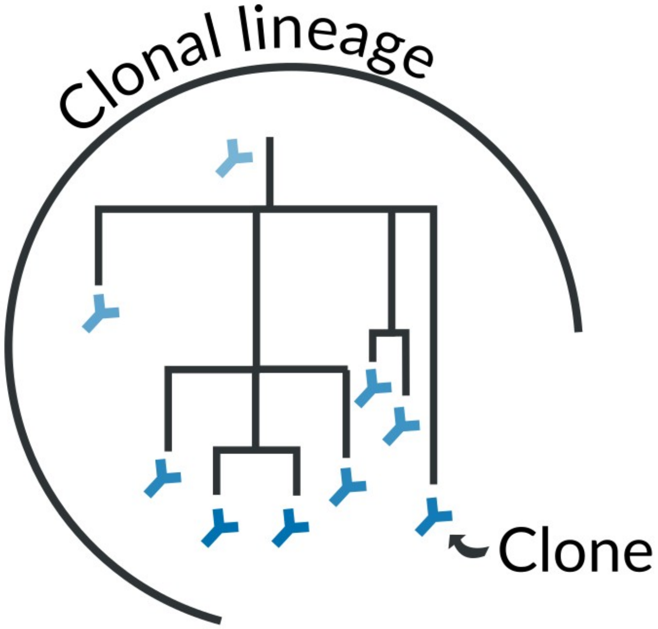
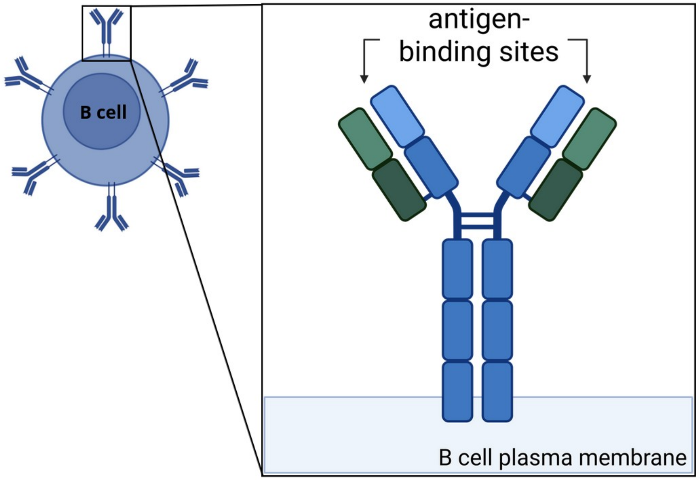


BCRs can evolve against antigens and acquire mutations

→ clonal reconstruction
→ phylogenetics

Protein Problems



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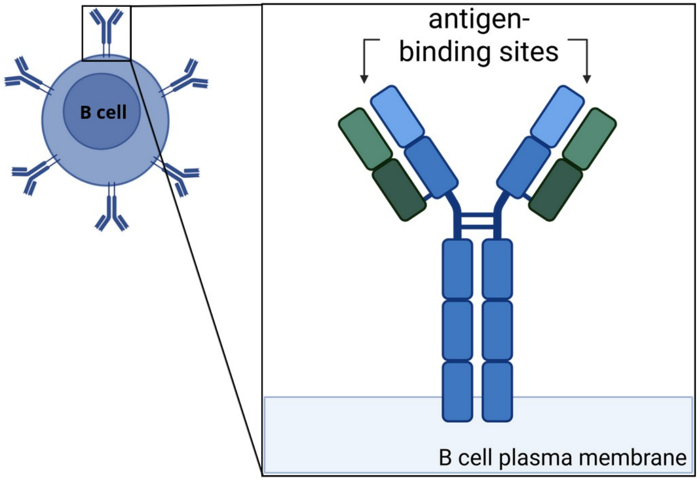
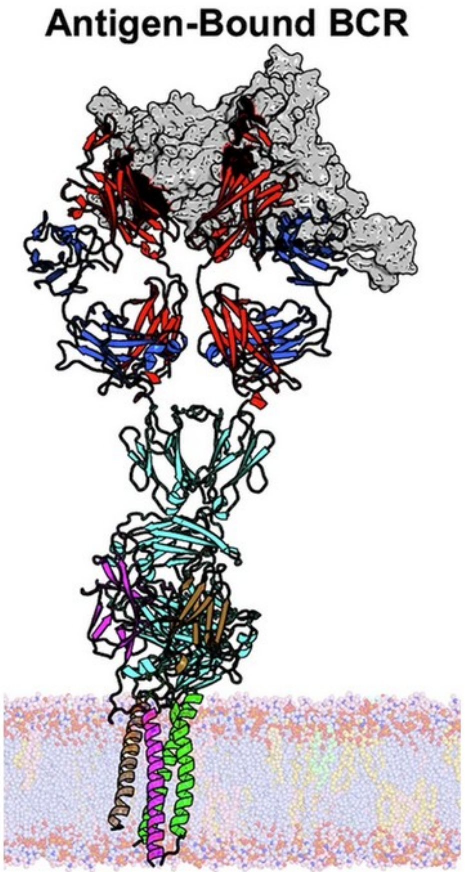
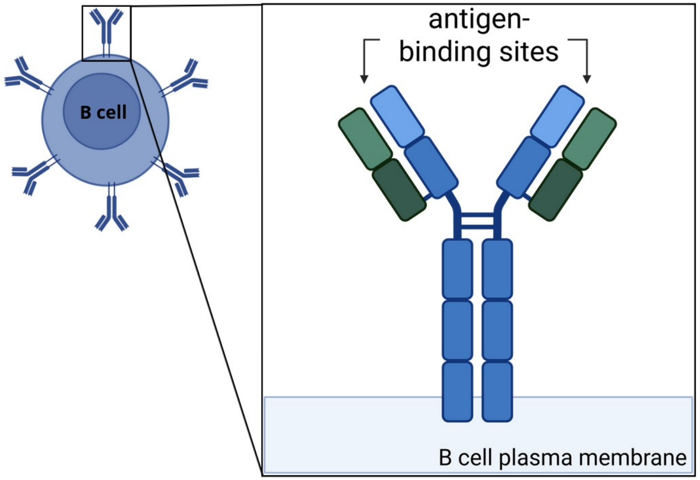
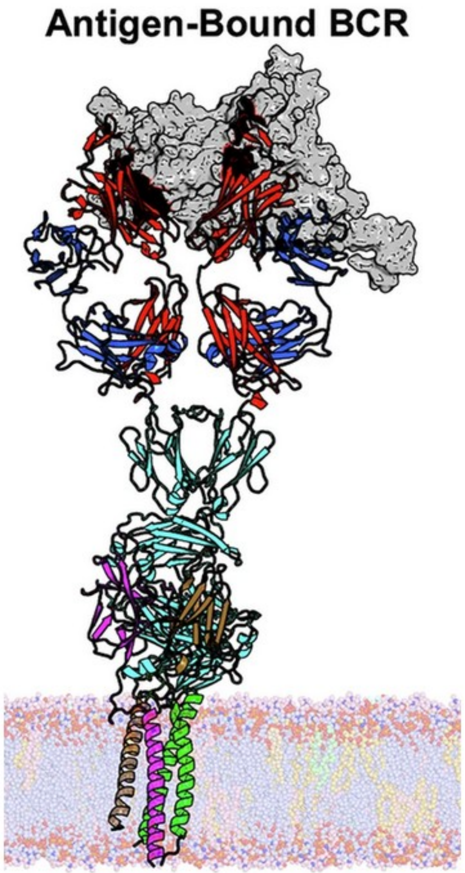




BCR / Antigen interaction

Protein Problems





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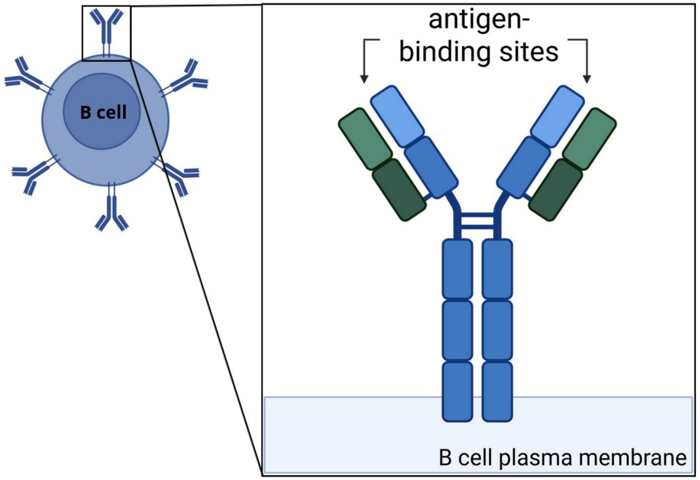

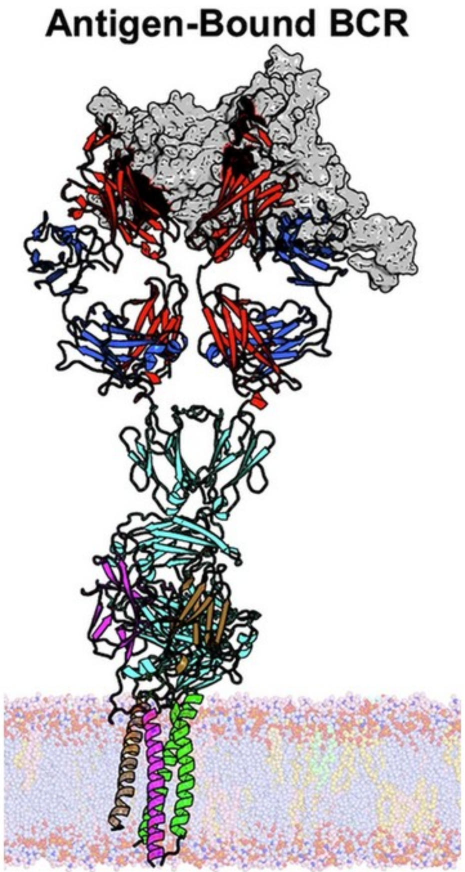

BCR / Antigen interaction

Prediction is near-impossible

Measurement is slow and expensive

Functional Characterization of the Lineages

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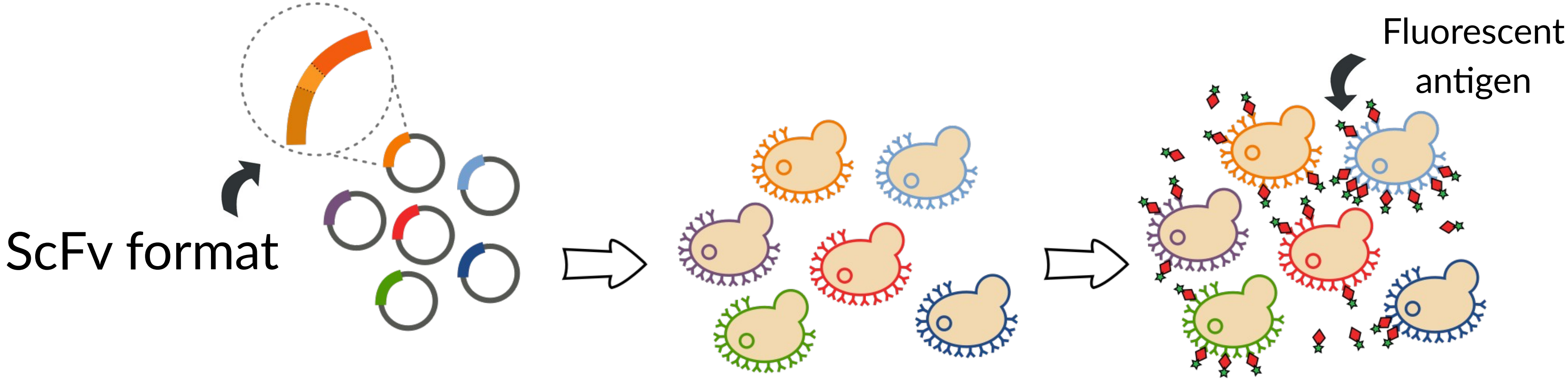
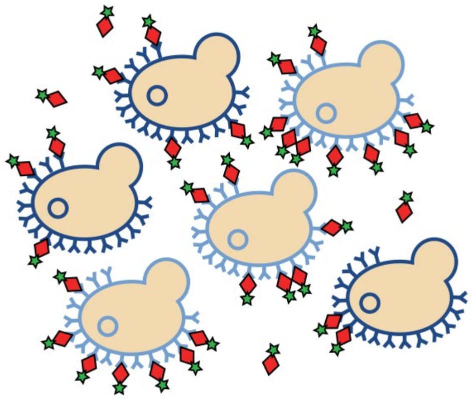
Yeast display expression of antibodies



Affinity Measurement in High-Throughput, Tite-Seq

Incubation at a fixed antigen concentration



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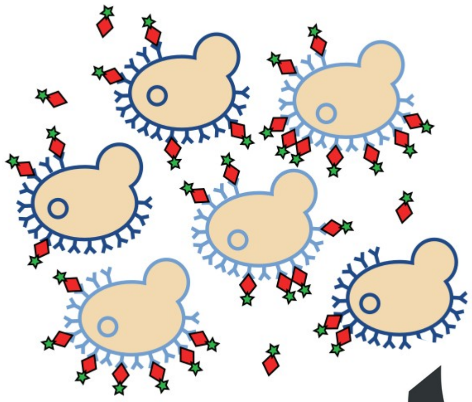
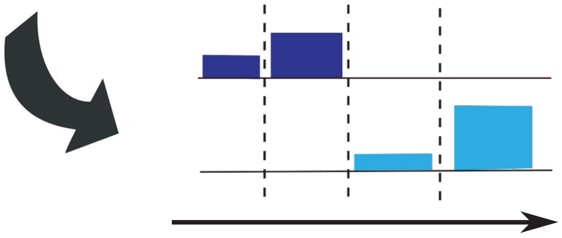


4-ways sort & sequencing



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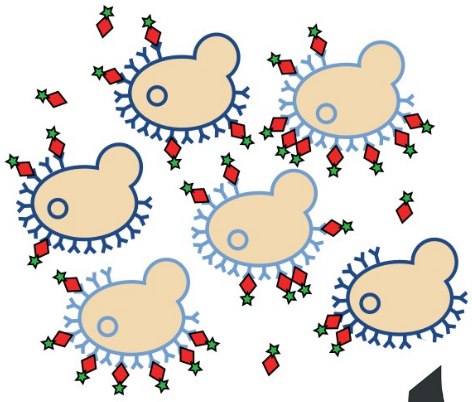

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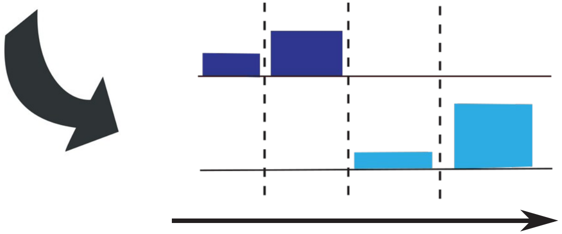
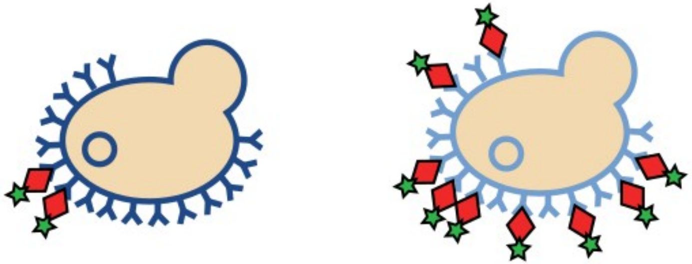


⇒ Estimate the average proportion of antigen bound to each distinct antibody



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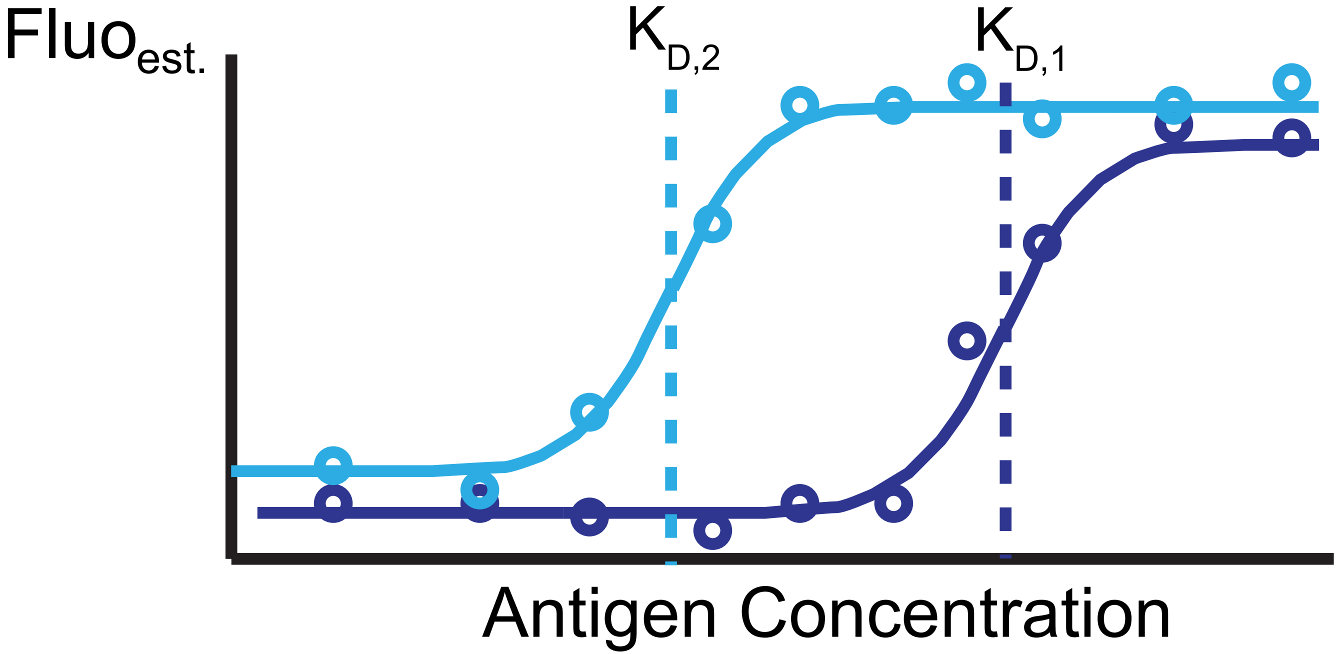
Repeat at different antigen concentrations to create the full titration curve.



Phillips, A. M. et al, eLife (2023)

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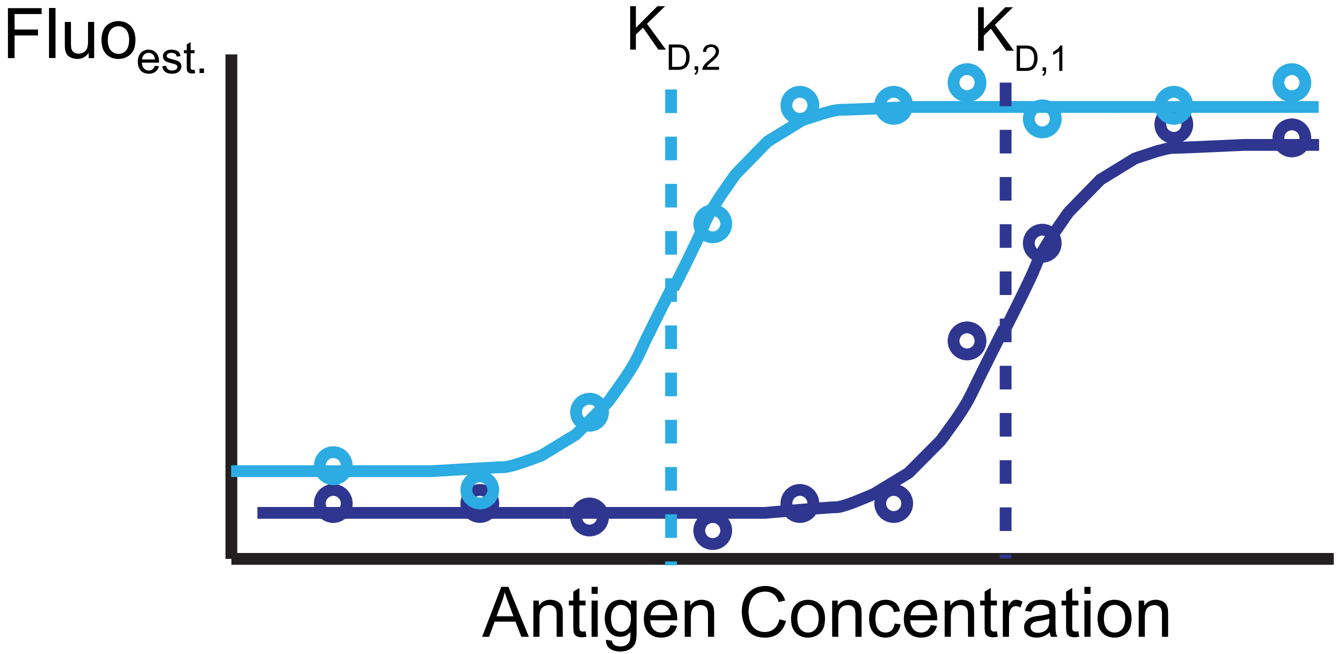
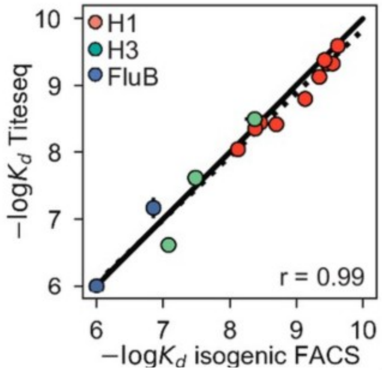
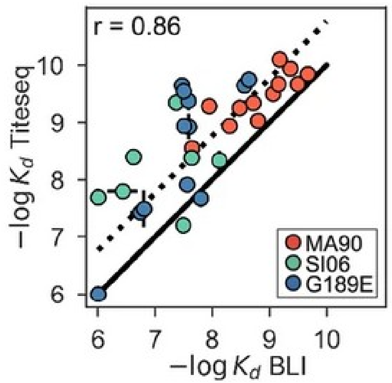




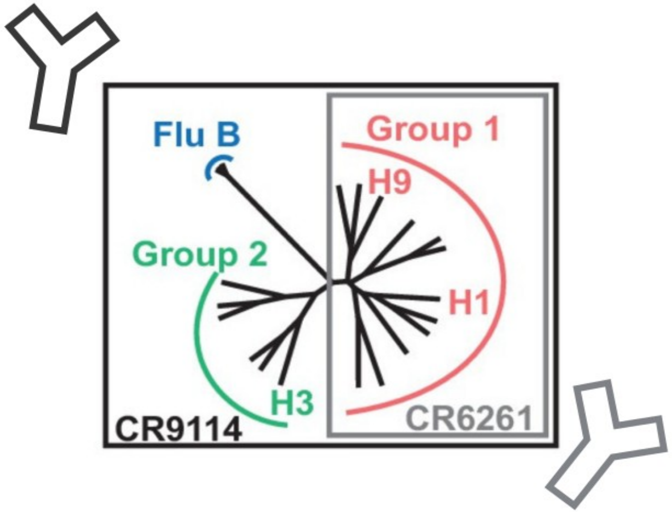
Versus titration



Versus Bio-Layer interferometry

Phillips, A. M. et al, eLife (2023)

Example: Evolution of Broadly Neutralizing Abs



We focused on two anti-influenza neutralizing antibodies **CR9114** and CR6261.

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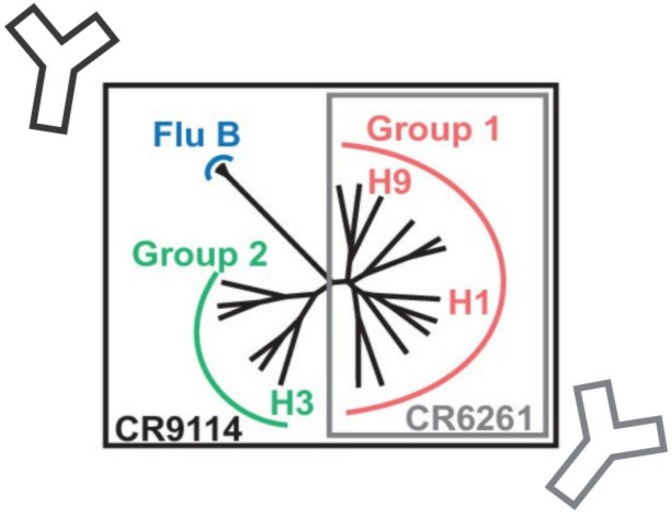

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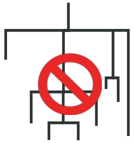

Throsby et al., Plos One (2008)
Dreyfus et al., Science (2012)

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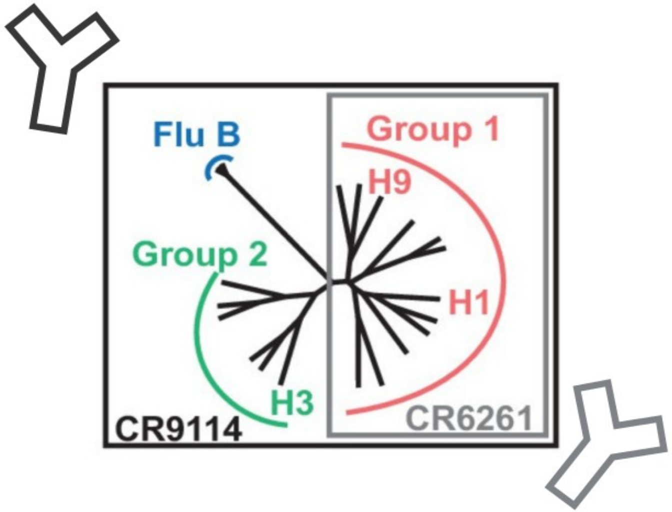

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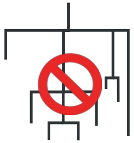

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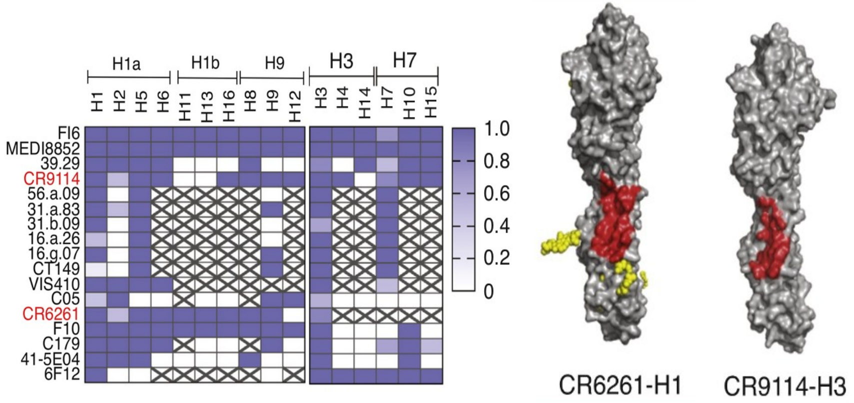

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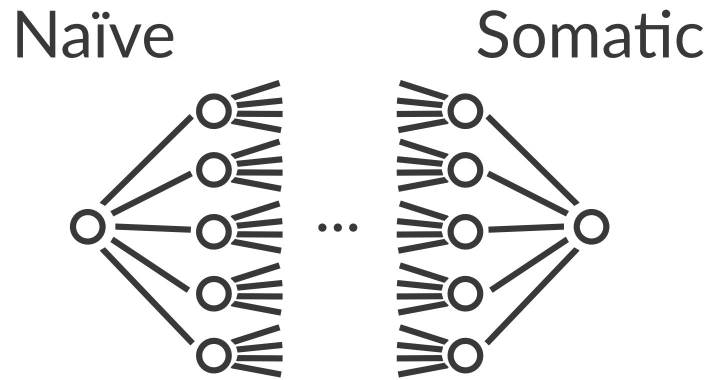
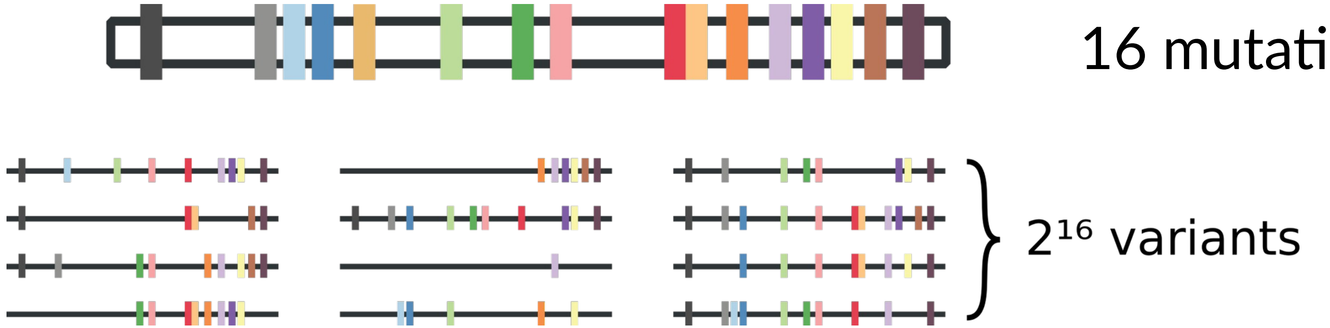


For both antibodies we recreated all possible intermediary genes between the naïve and the somatic.



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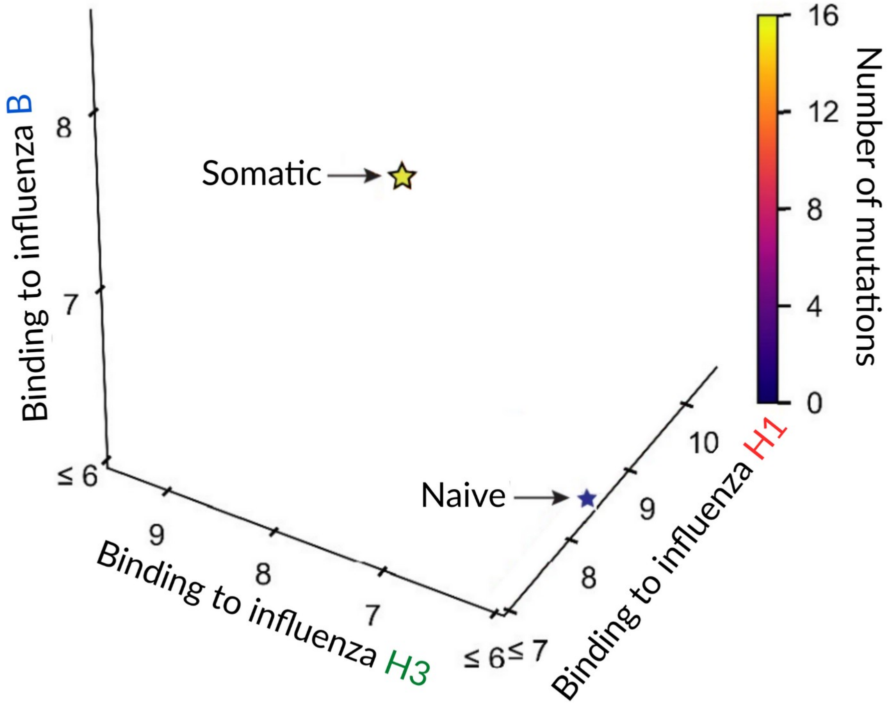
The somatic antibody binds all 3 antigens and the naive one only H1.



Phillips, A. M. et al. elife (2021)

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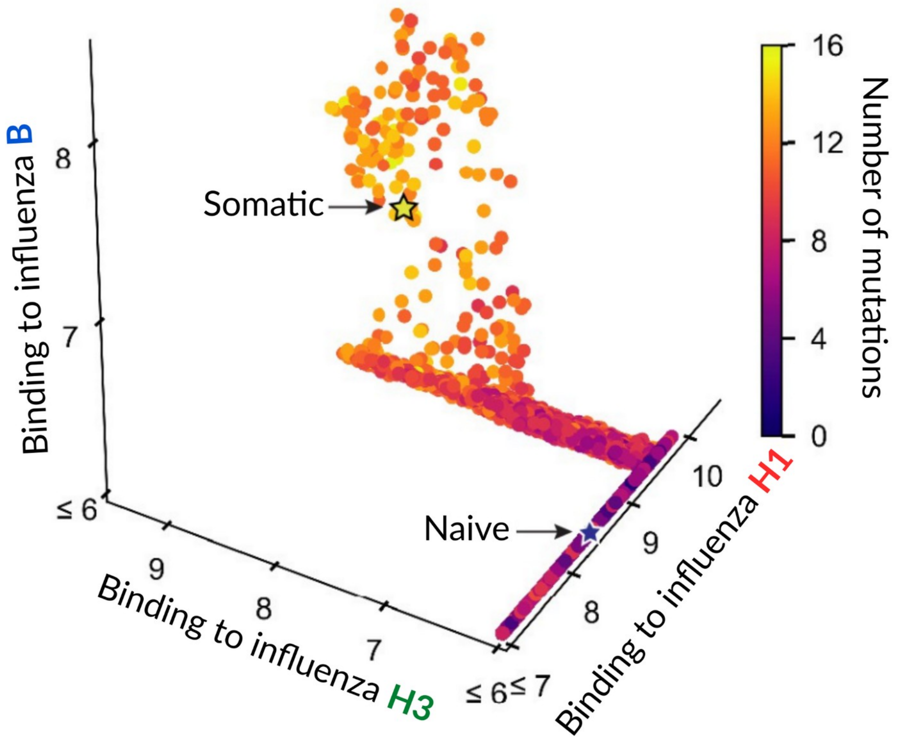

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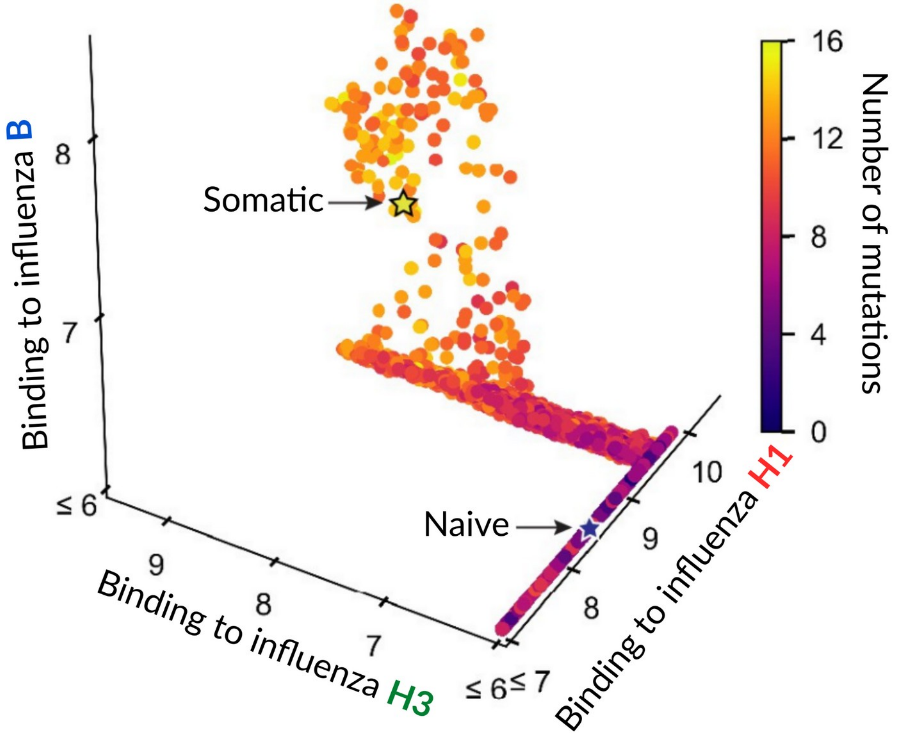
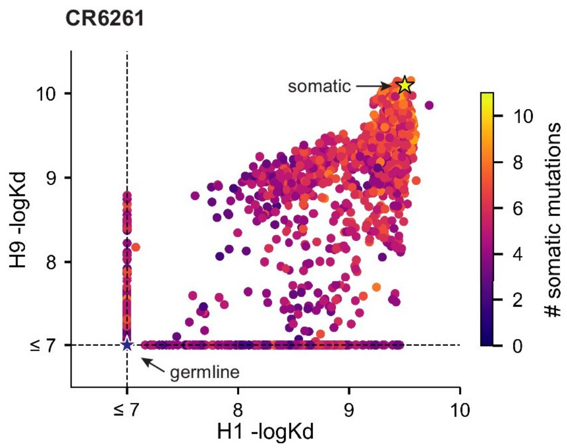
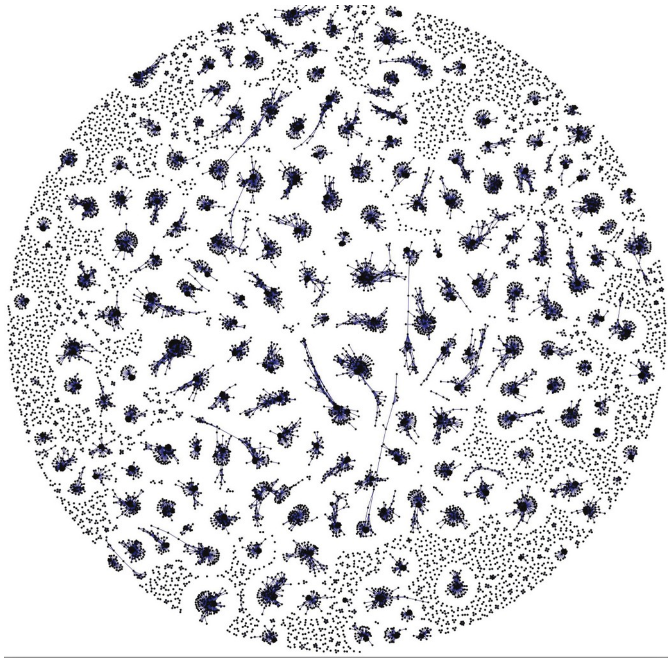




Not true for **CR6261**

Phillips, A. M. et al. elife (2021)

From one antibody to the repertoire



Real responses are multi-clonal

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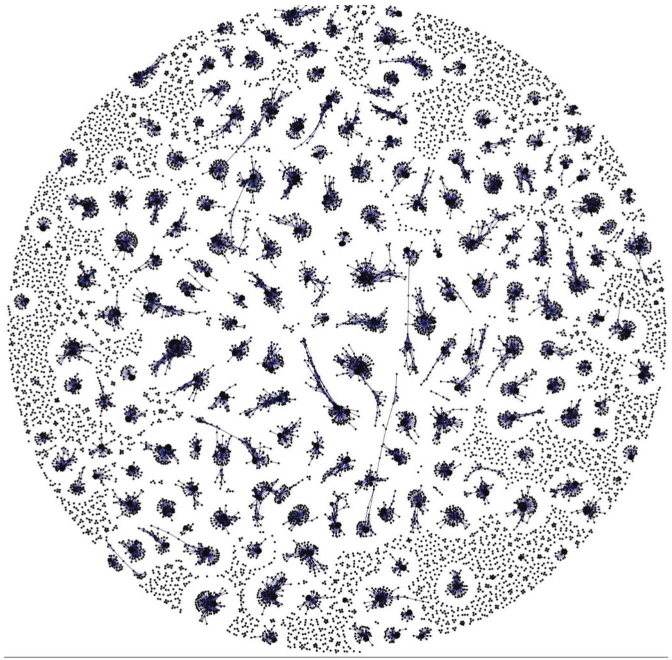

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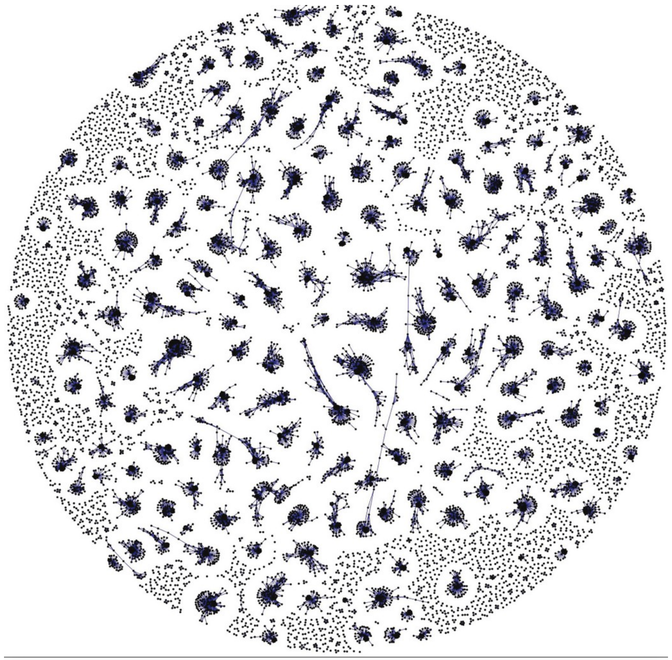

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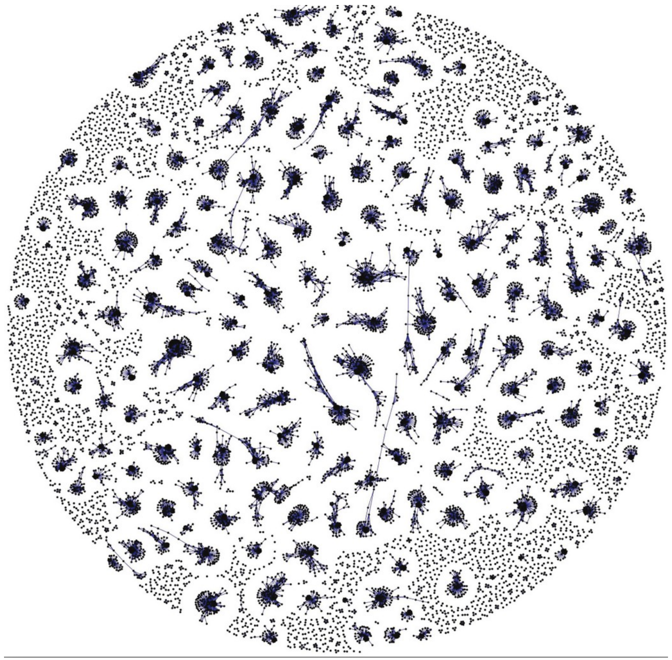

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What are their properties ?

Can we detect them computationally ?

Antibody response during Vaccination

Vaccination as a model for studying the repertoire response: known antigens & known timecourse.

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Dynamics of B cell repertoires and emergence of cross-reactive responses in patients with different severities of COVID-19

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Antibody response during Vaccination

Ciml IMMUNOLOGY

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Longitudinal analysis of the peripheral B cell repertoire reveals unique effects of immunization with a new influenza virus strain

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PBMC B-cell response to SARS-CoV-2 vaccination

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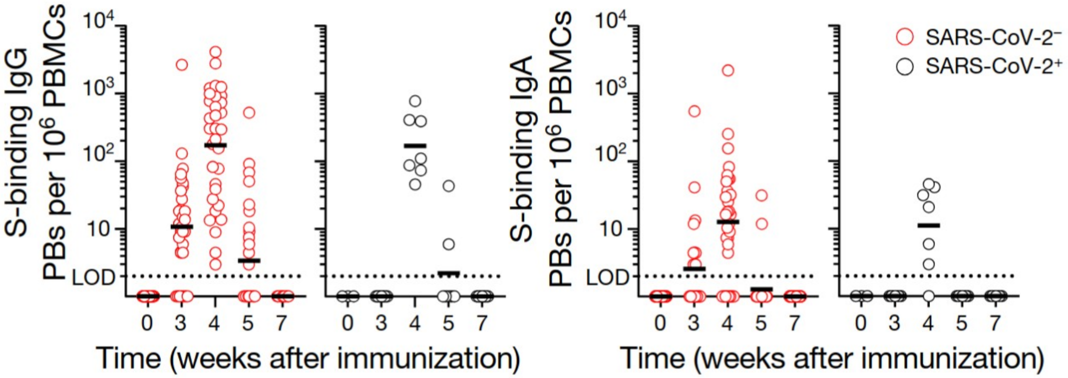

Turner, J. S. et al. Nature (2021)

Short term (< 6 weeks): Plasmablast and antibody production

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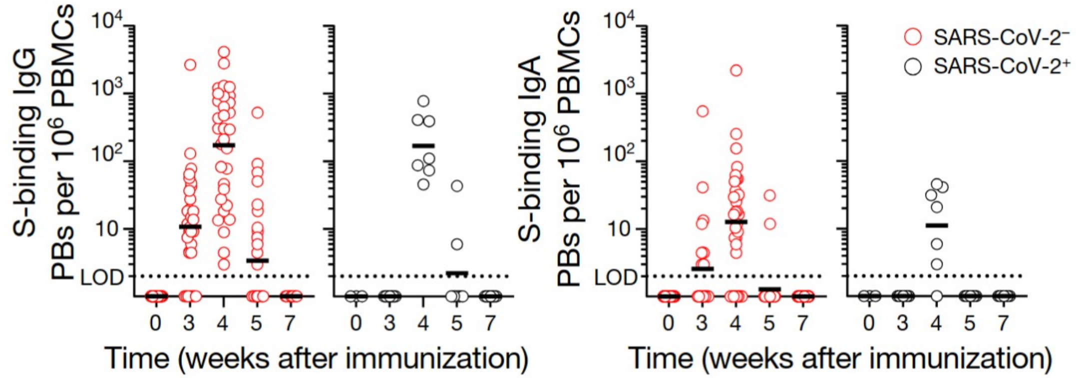

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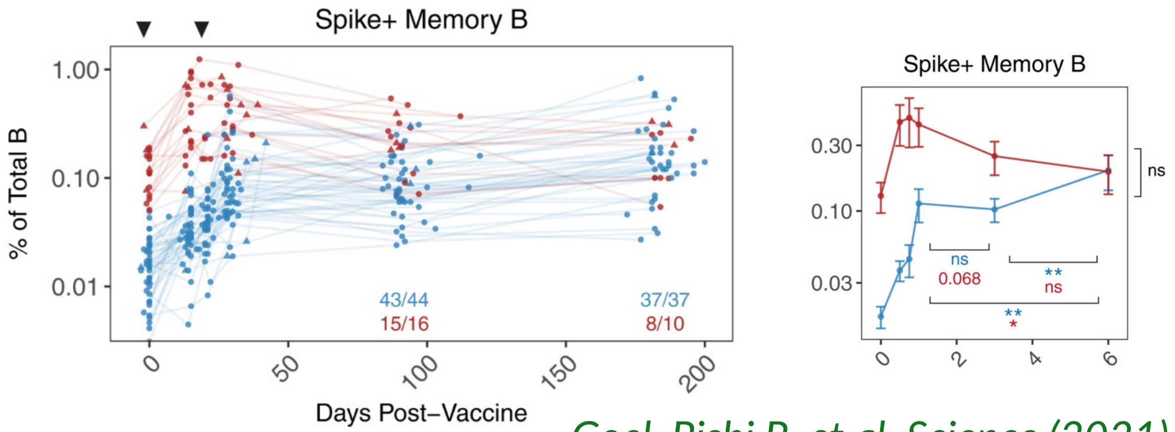
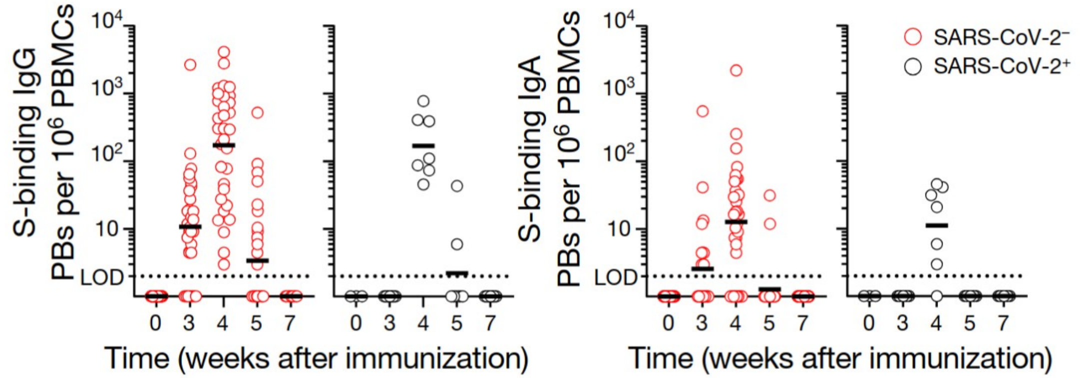


Goel, Rishi R. et al. Science (2021)

Long term: Memory B-cells

PBMC B-cell response to SARS-CoV-2 vaccination



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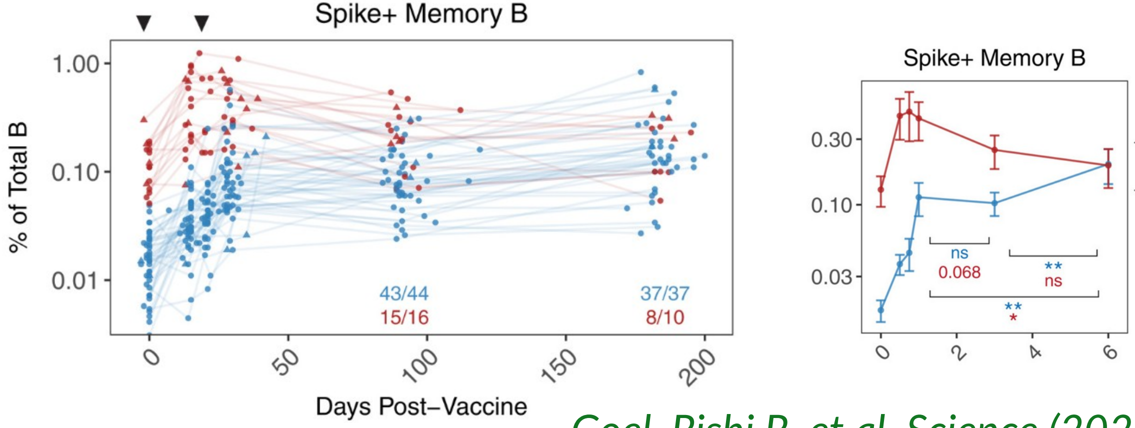
Short term (< 6 weeks): Plasmablast and antibody production



Goel, Rishi R. et al. Science (2021)

Long term:
Memory B-cells

0.1-1% of the total B-cells are spike+

Antibody response during Vaccination

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

Antibody response during Vaccination

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

1. Long-term response

Antibody response during Vaccination

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

1. Long-term response

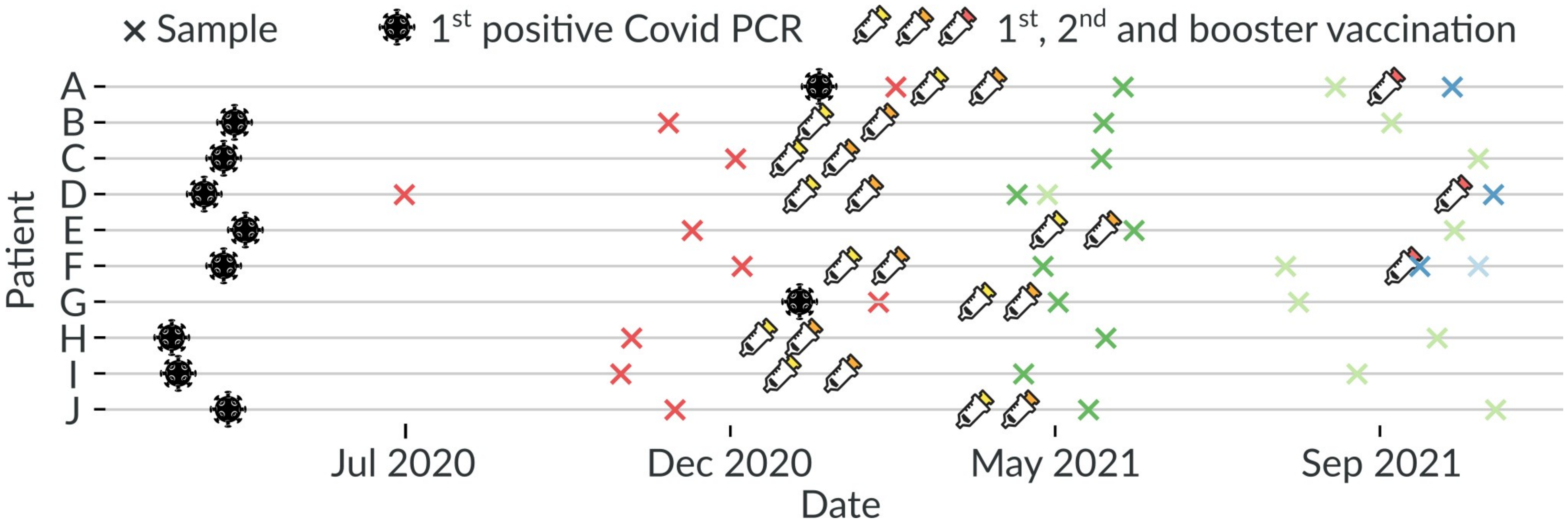
2. High depth

Antibody response during Vaccination

Following the evolution of the BCR repertoire alongside SARS-CoV-2 vaccination.

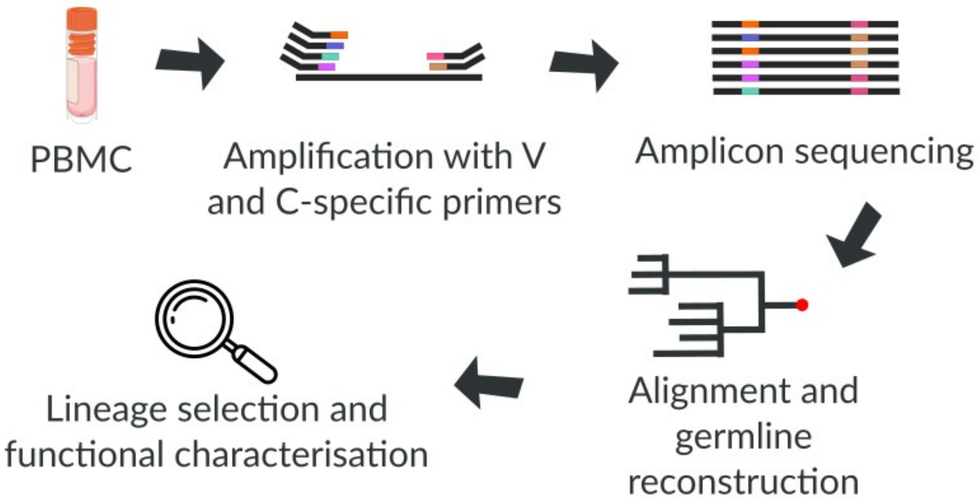
1. Long-term response
2. High depth
3. Functional characterization

Samples timeline



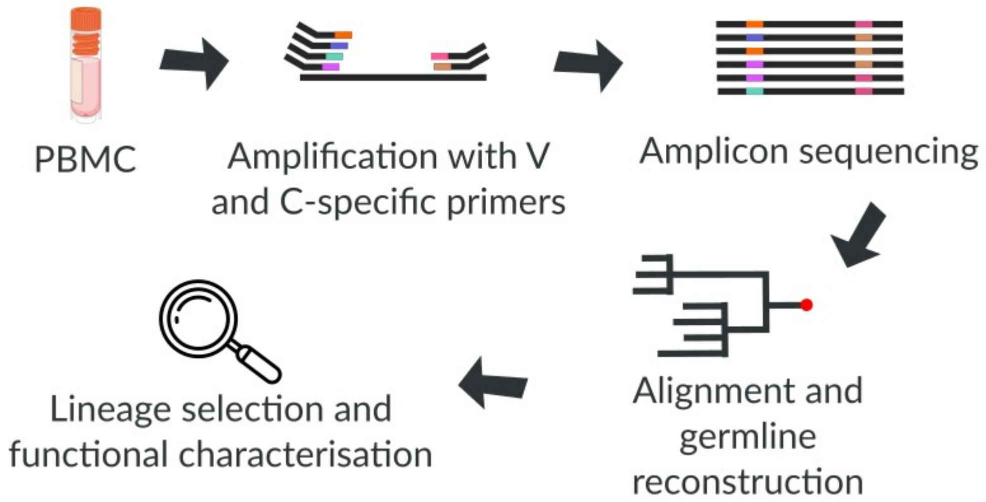
Longitudinal sampling on 10 individuals. Collaboration with the Yu lab at the Ragon institute, 10^9 B cellls.

Sequencing



Deep bulk repertoire sequencing of heavy and light chains, 34 samples × 5 replicates.

Sequencing



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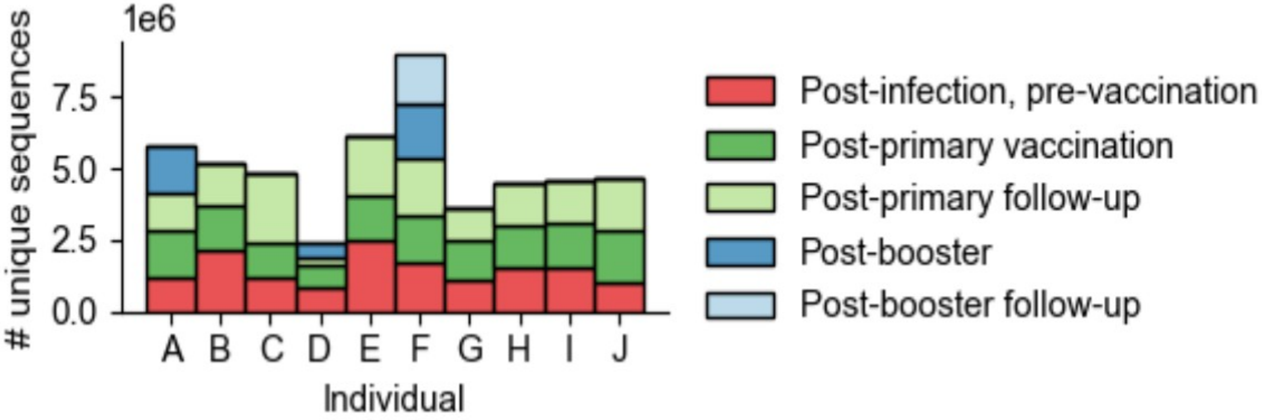
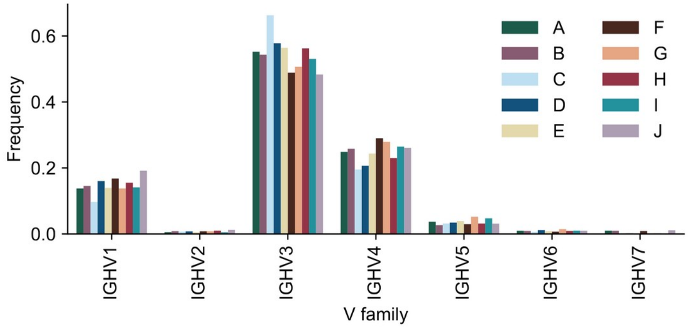
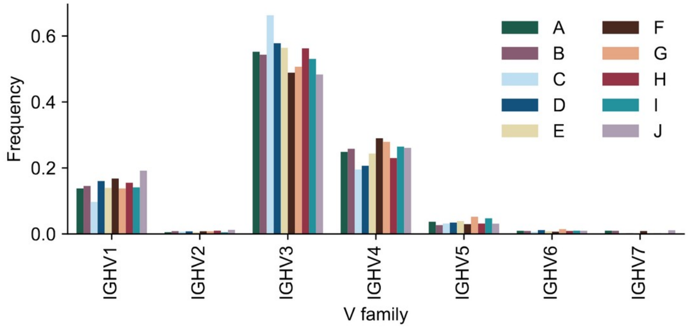


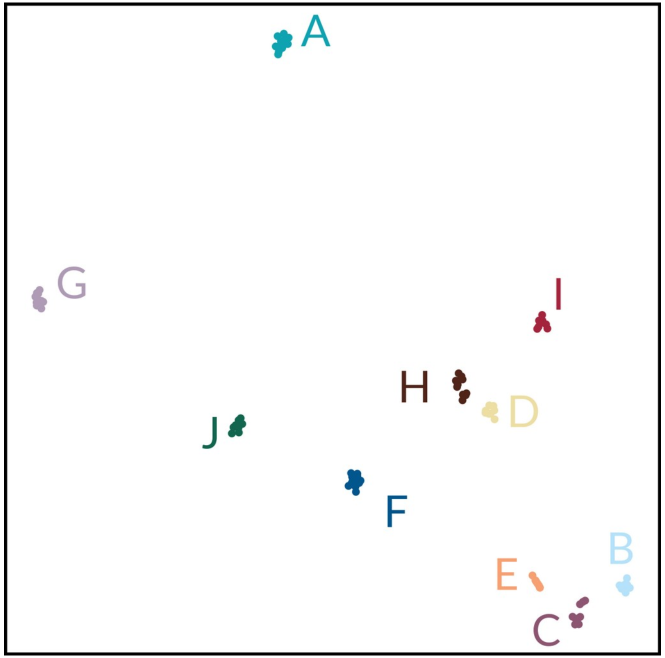
The Immune Repertoire



Repertoire change a lot but their statistical property don’t.

The Immune Repertoire





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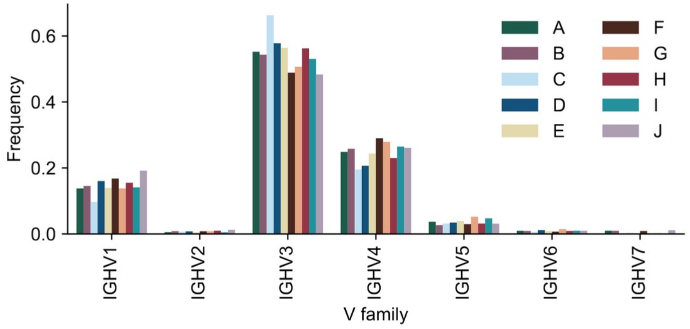

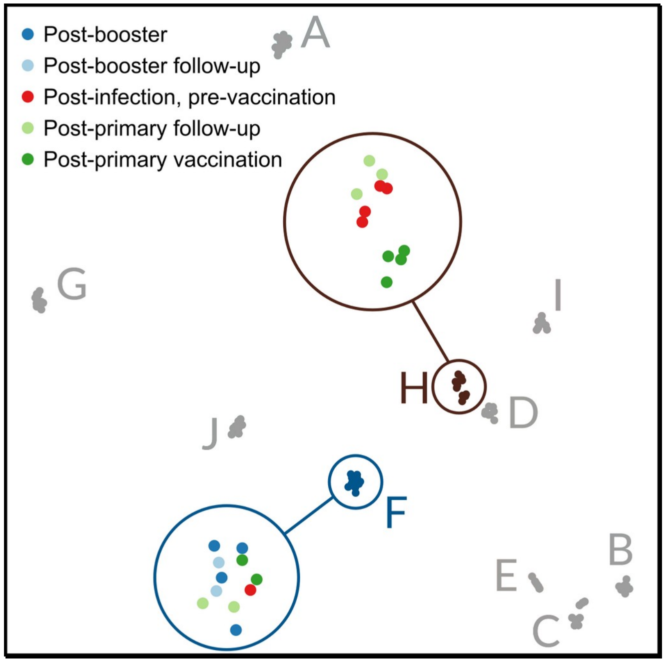

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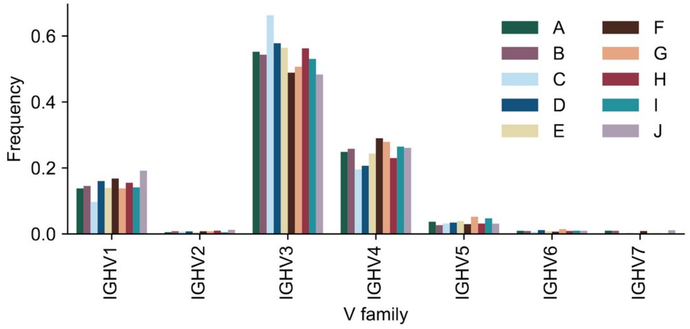

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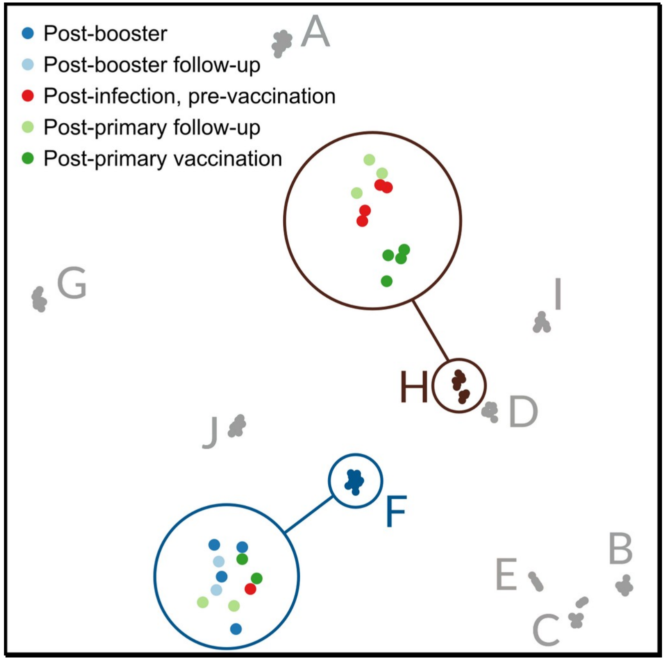
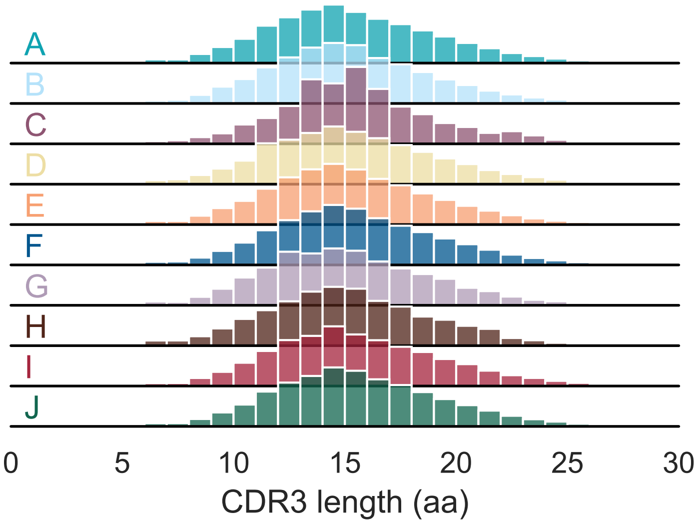
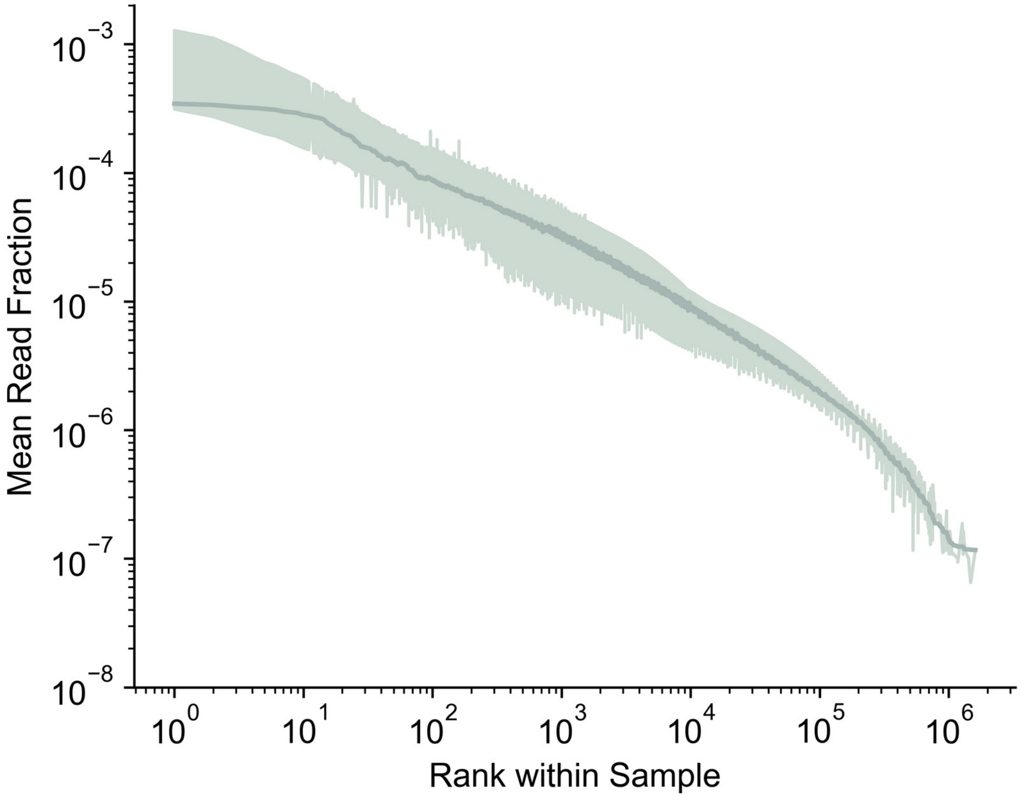




Making a library

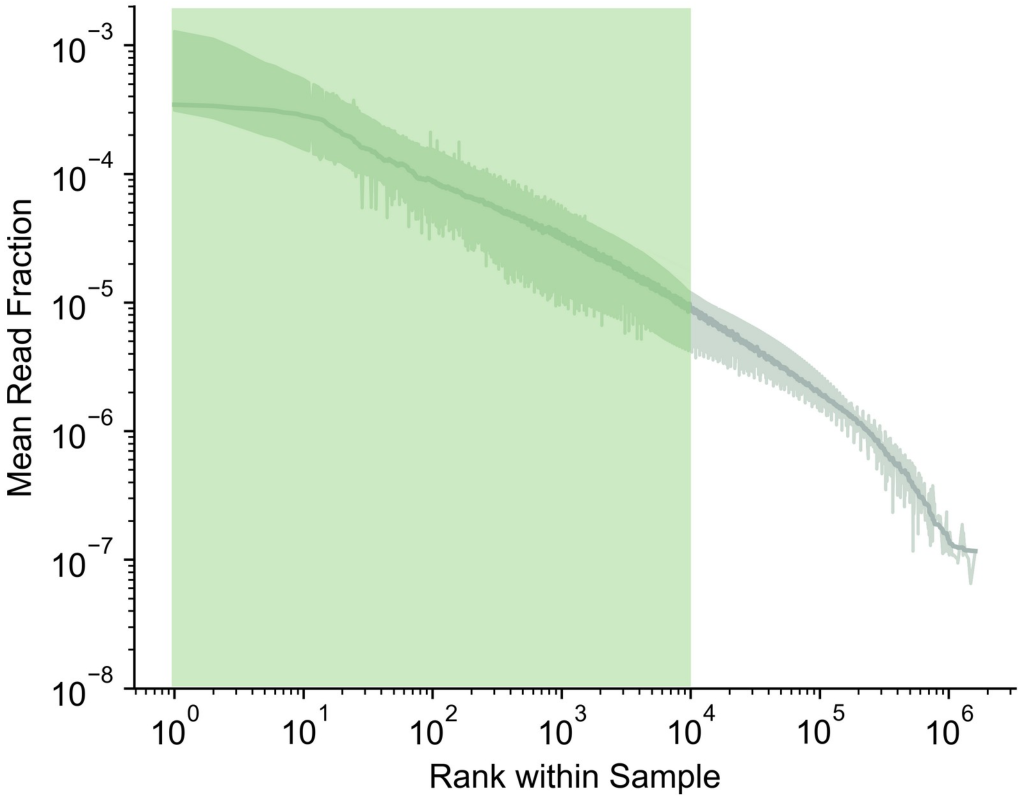
BCR frequencies follow "power-law" distribution



Making a library

BCR frequencies follow "power-law" distribution

We focus on the **10'000 largest clonal lineages.**

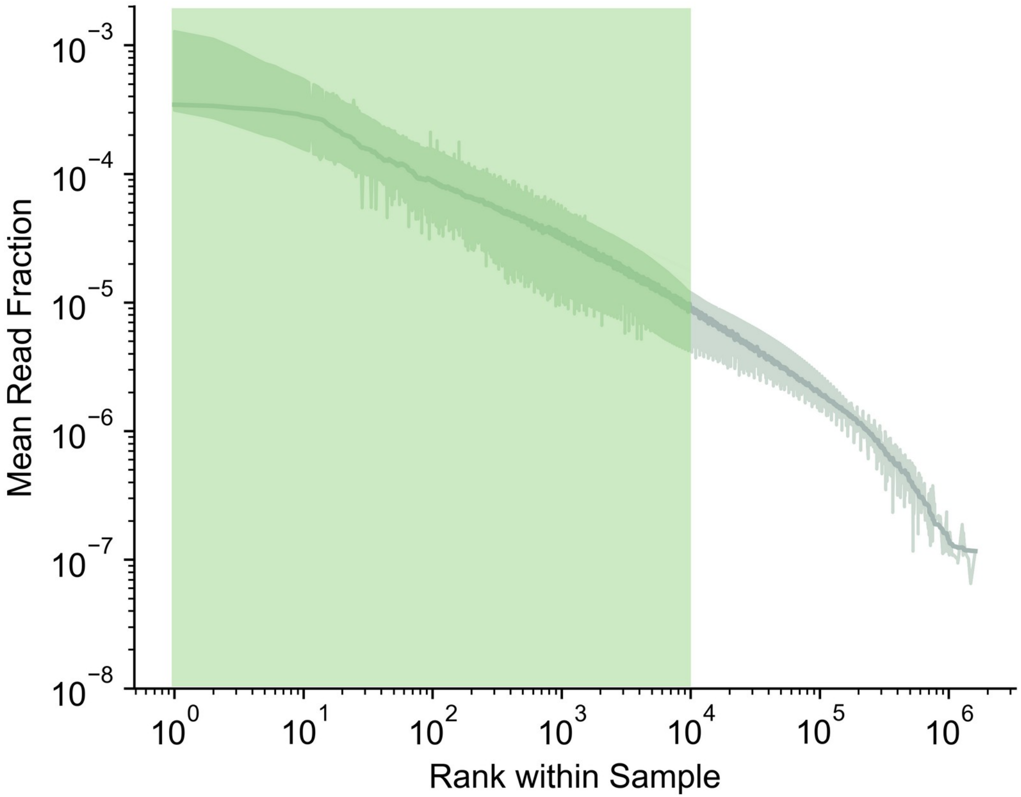


Making a library

BCR frequencies follow "power-law" distribution

We focus on the **10'000 largest clonal lineages.**

(still a very small part of the overall repertoire ~ 0.1%)



Making a library

Bulk sequencing limitation: **no heavy-light pairing**

Making a library

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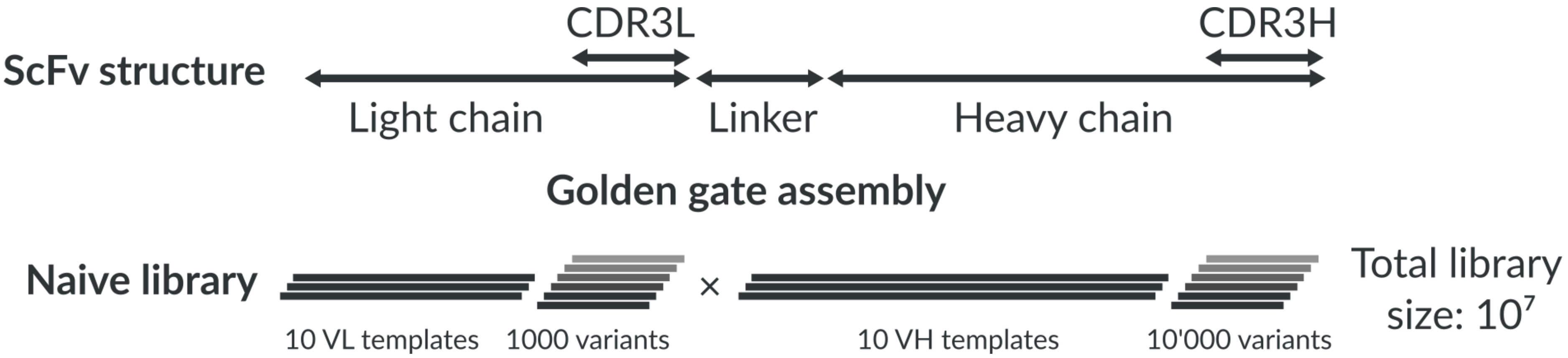
⇒ Combinatorial approach

Making a library

Bulk sequencing limitation: **no heavy-light pairing**

⇒ Combinatorial approach

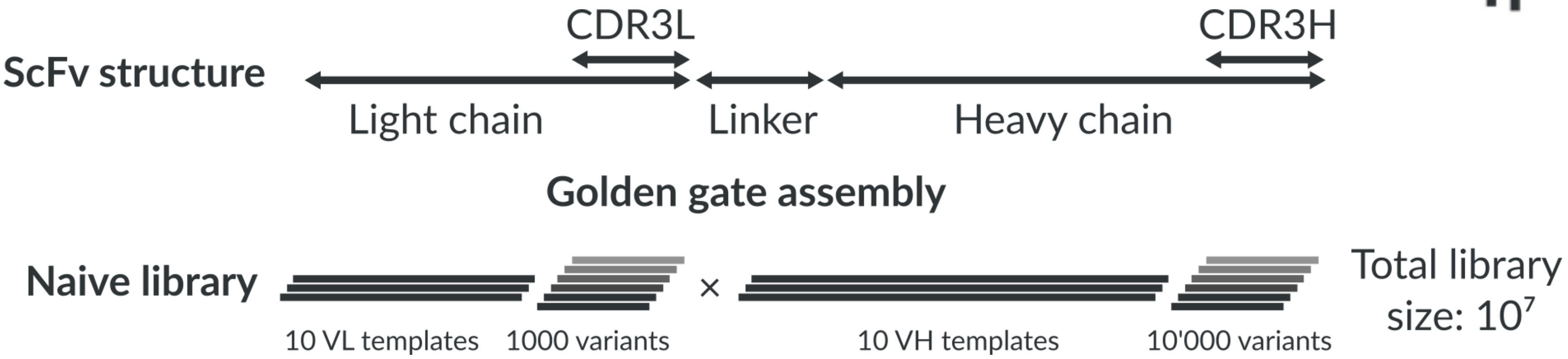


Making a library

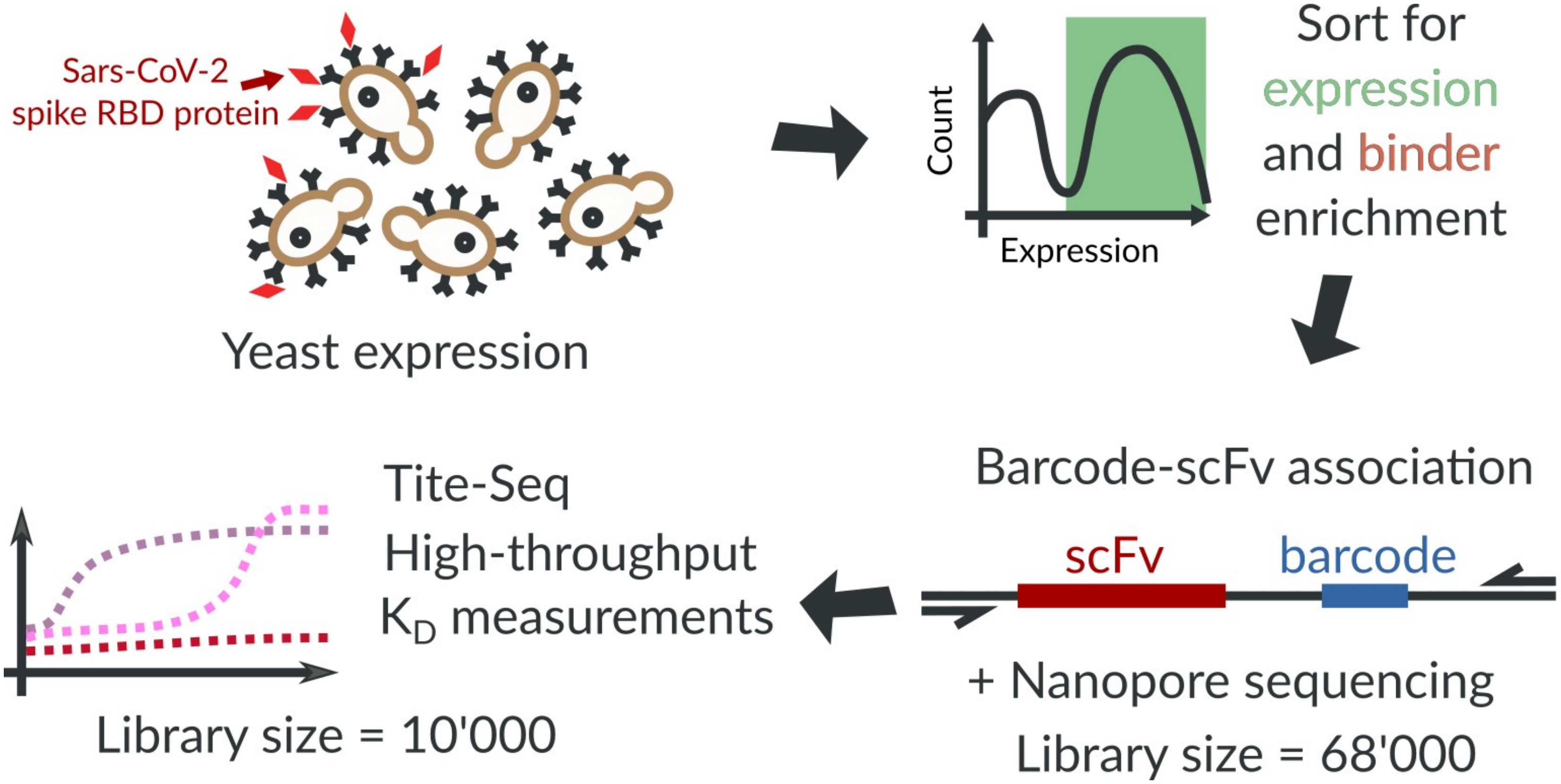
Bulk sequencing limitation: **no heavy-light pairing**

⇒ Combinatorial approach

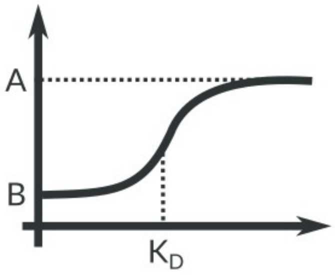
Focus on the reconstructed naive sequence



Functional characterization

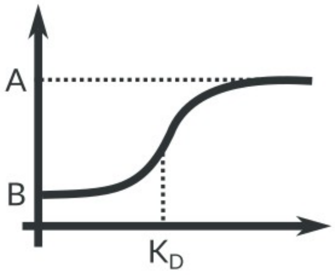


Binders



Differentiate between binders and non-binders

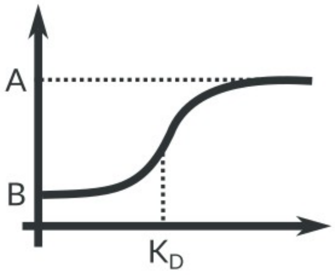
Binders



Differentiate between binders and non-binders

2% of the repertoire binds

Binders

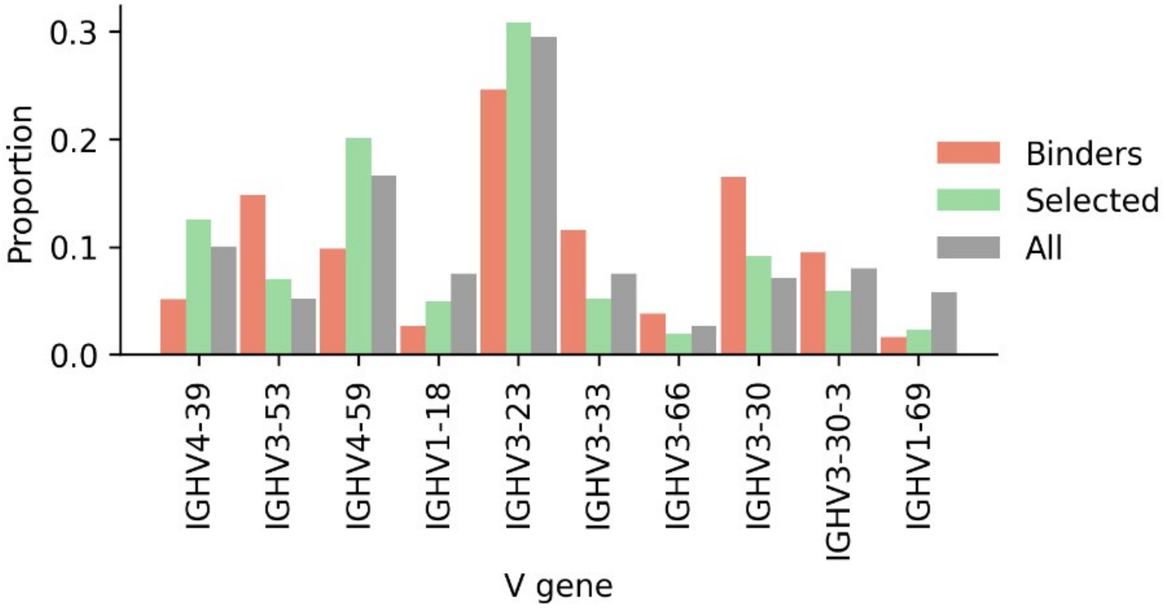


Differentiate between binders and non-binders

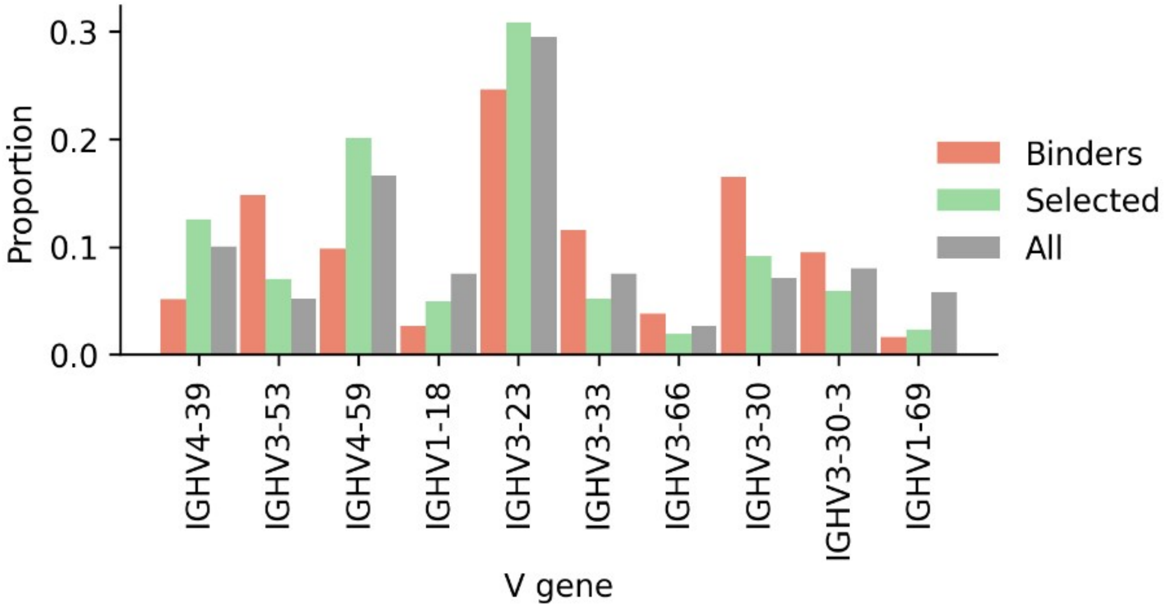
Measured K_D

2% of the repertoire binds

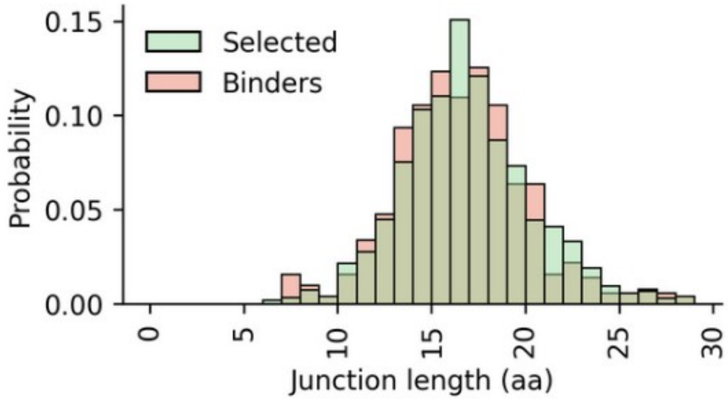
A diverse response



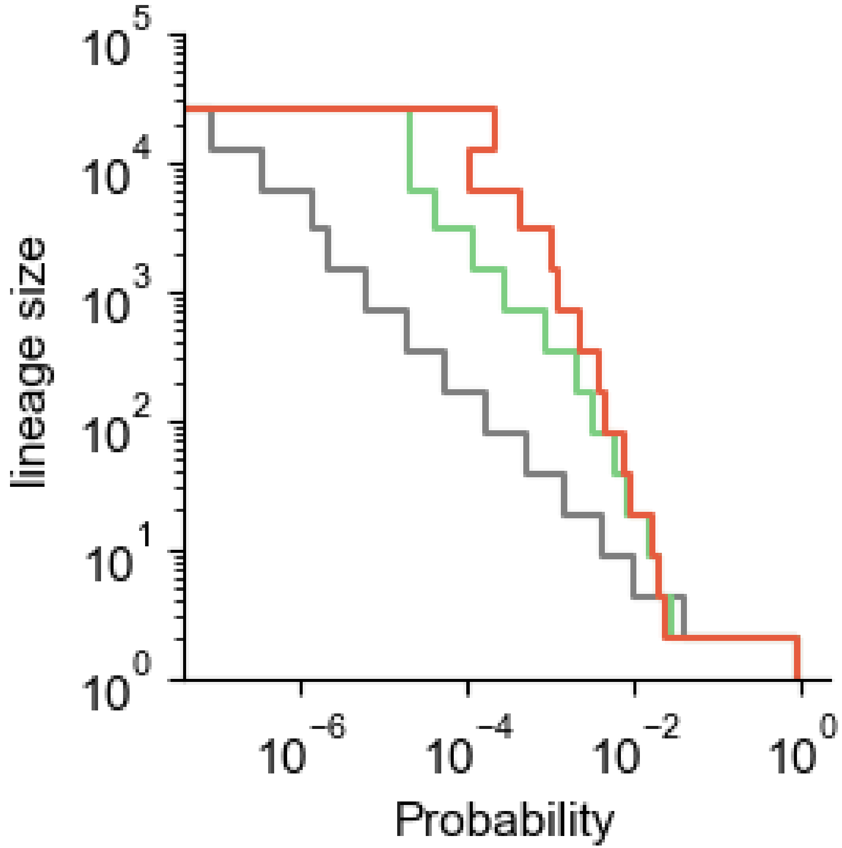
A diverse response



Binder lineages come from all V-genes (that we tested) and have diverse CDR3 lengths

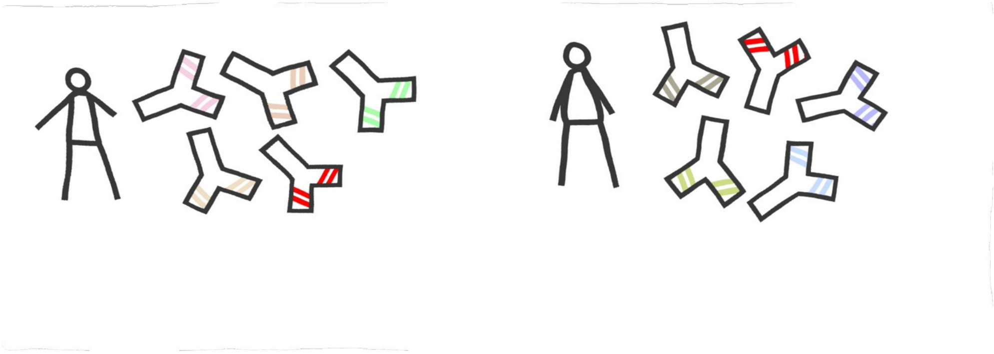


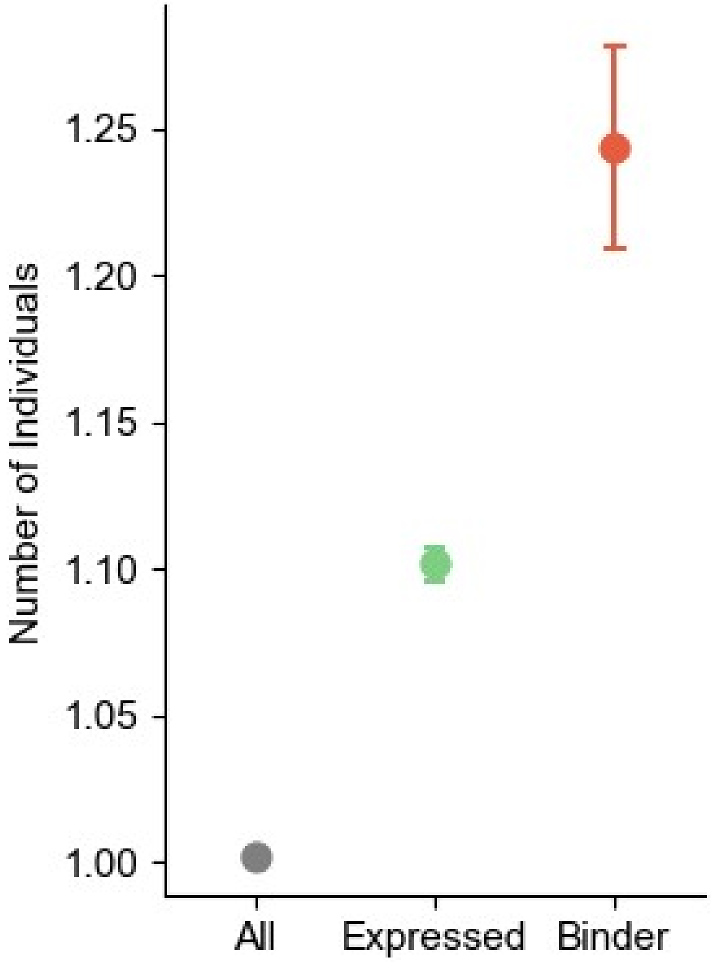
Binder lineages are larger



Binder lineages are more shared between individuals

Binder lineages are more likely to be "public" among vaccinated individuals





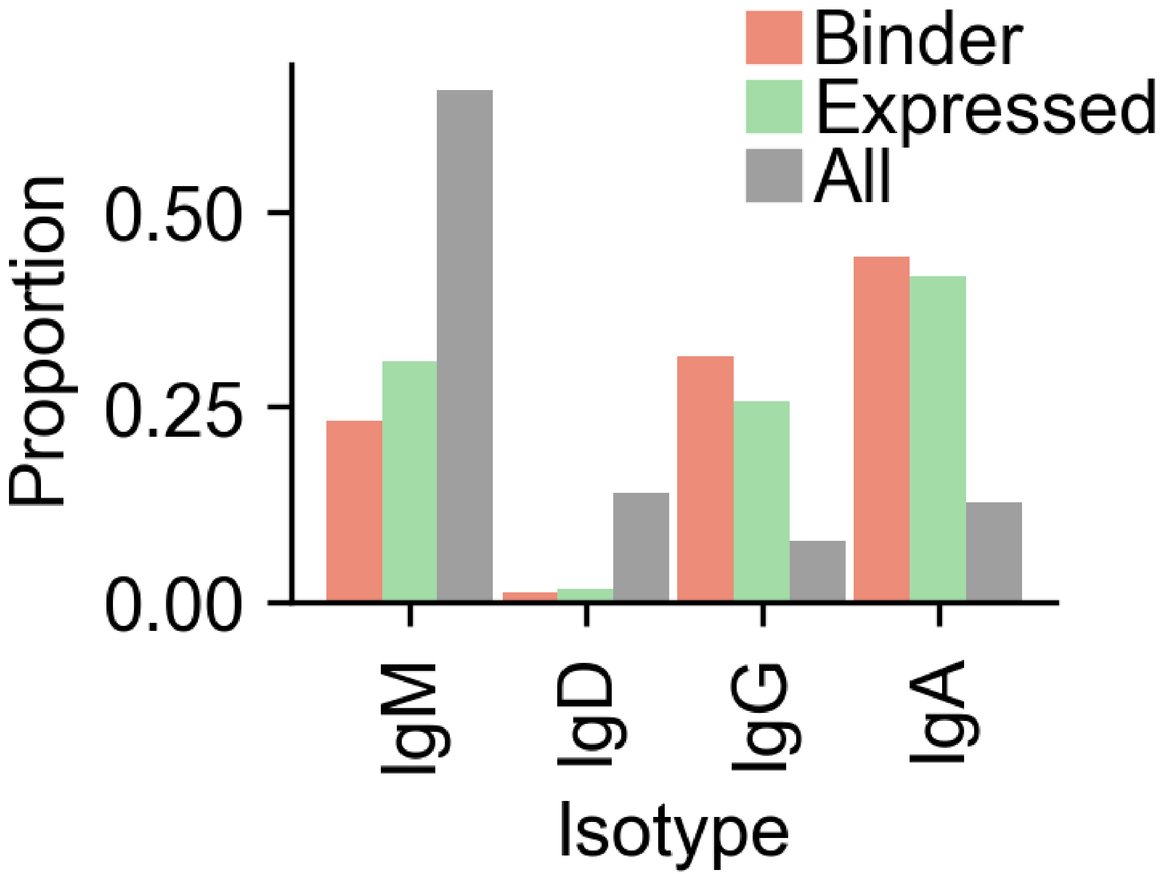
Binder lineages evolve more

Lineage with binding germlines gain new sequences

Binder lineages evolve more

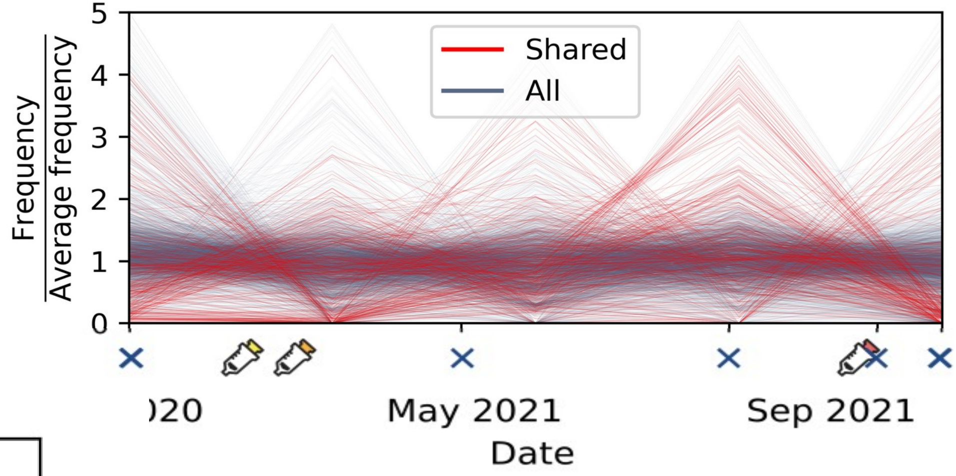
Lineage with binding germlines gain new sequences

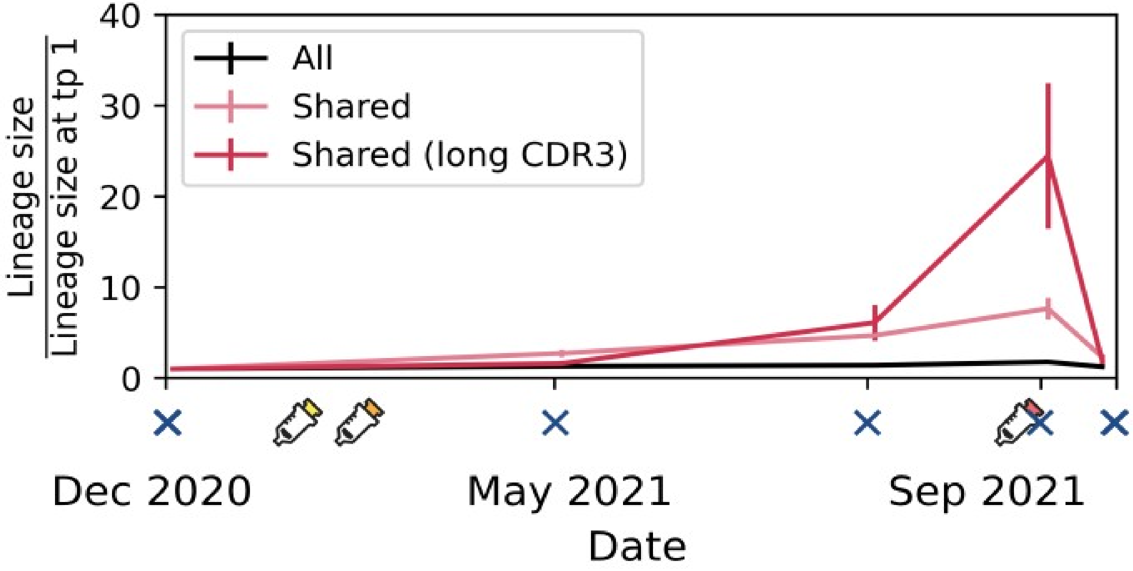
Binder lineages are more often class-switched



These different metrics are not independent

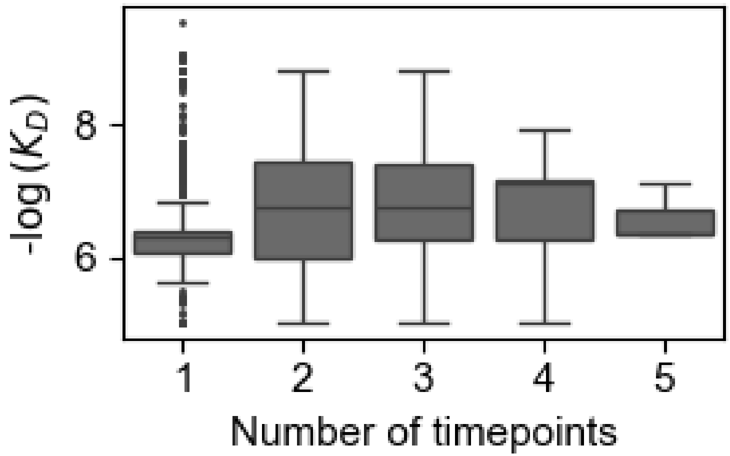
Shared lineages are more active





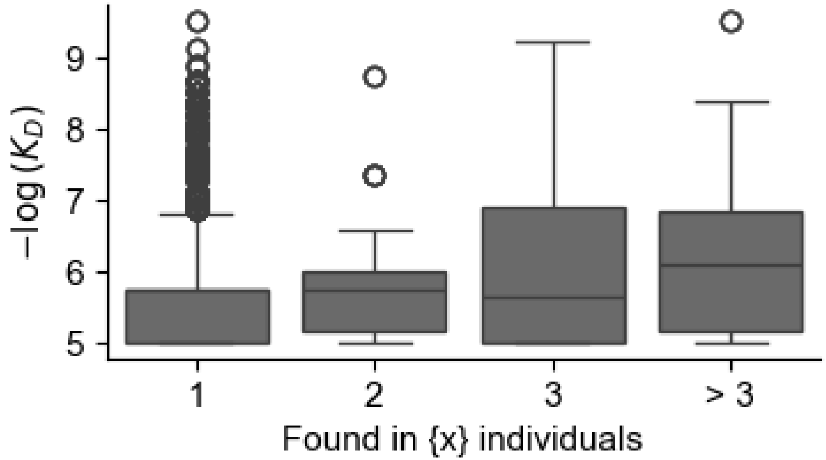
On average, shared lineages increase in size over time.

The affinity of the germline sequence matters too !



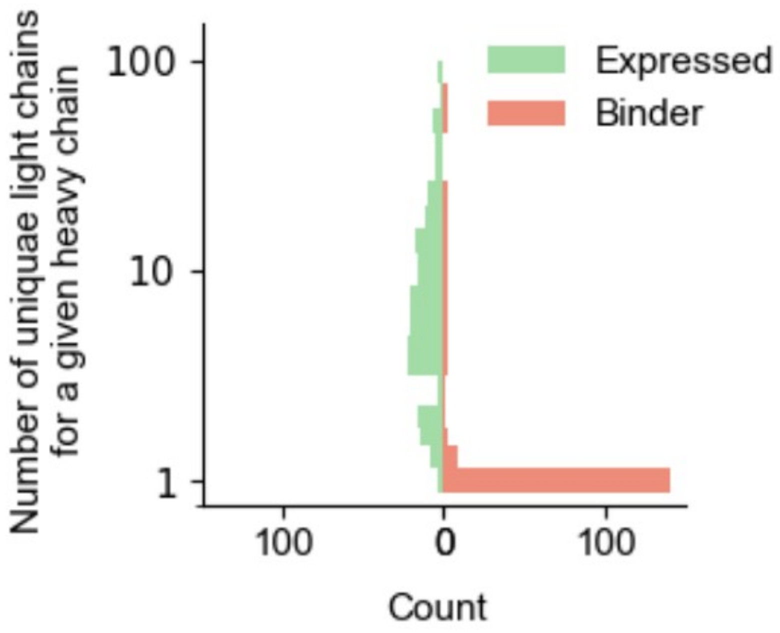
High affinity germline are found across more timepoints

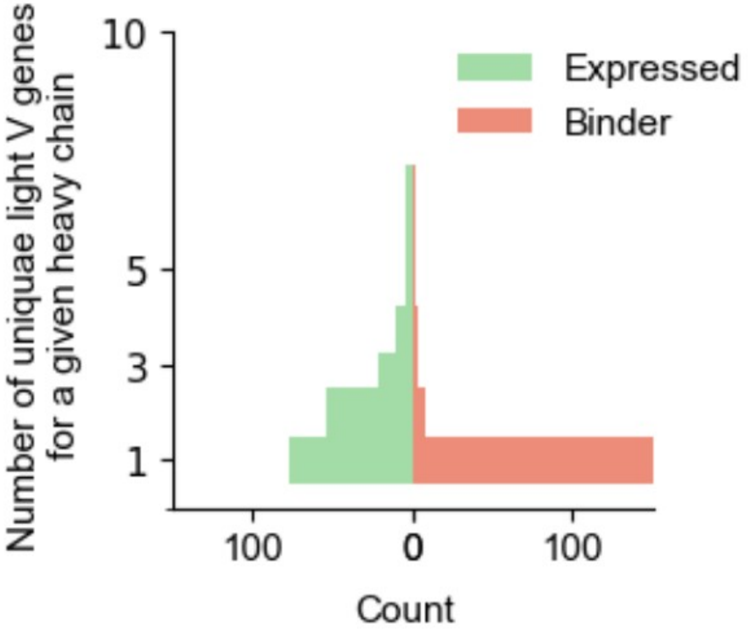
High affinity germline are more shared



Structural properties

The light chain is crucial for specificity



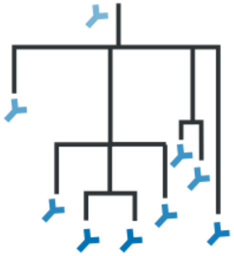


Structural properties

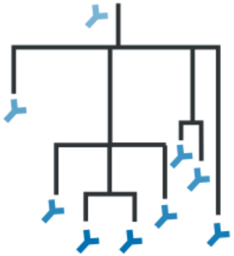
Alphafold is not doing very well here but is showing promise.

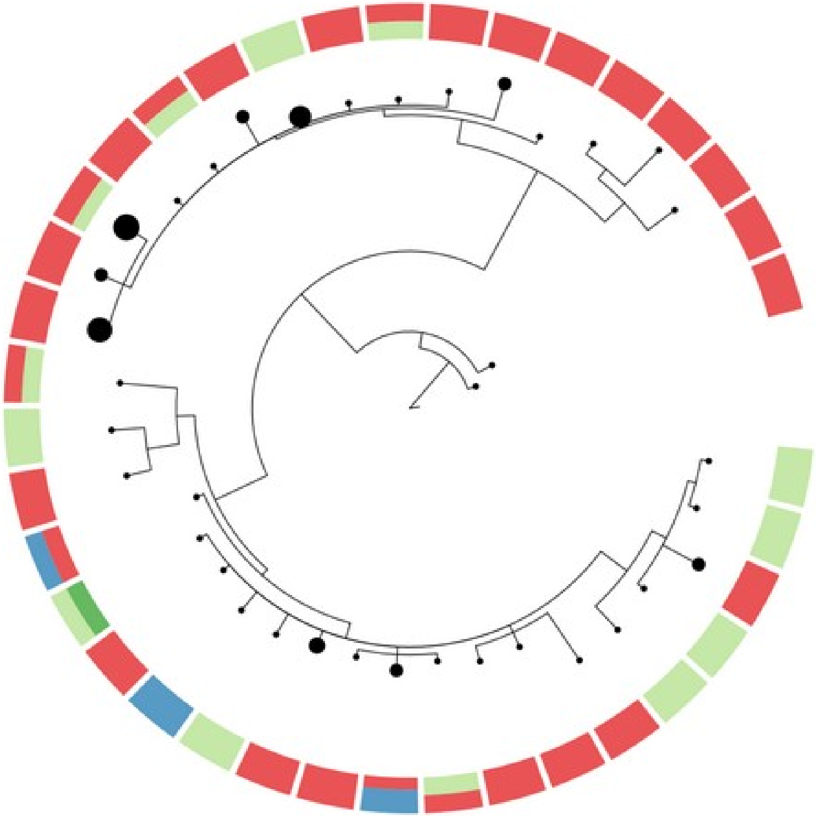
UMAP based on Alphafold internal representation. Circled sequences are potential predicted binders.

Lineage evolution

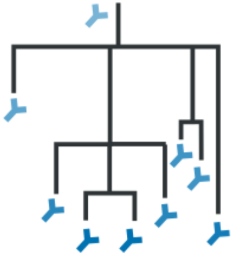


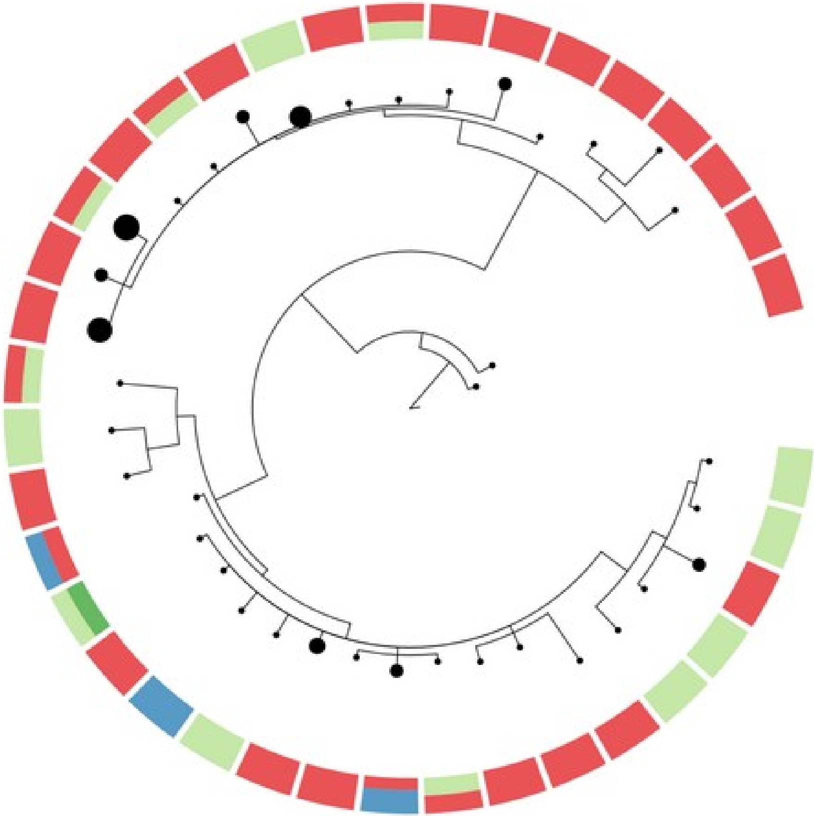
Lineage evolution



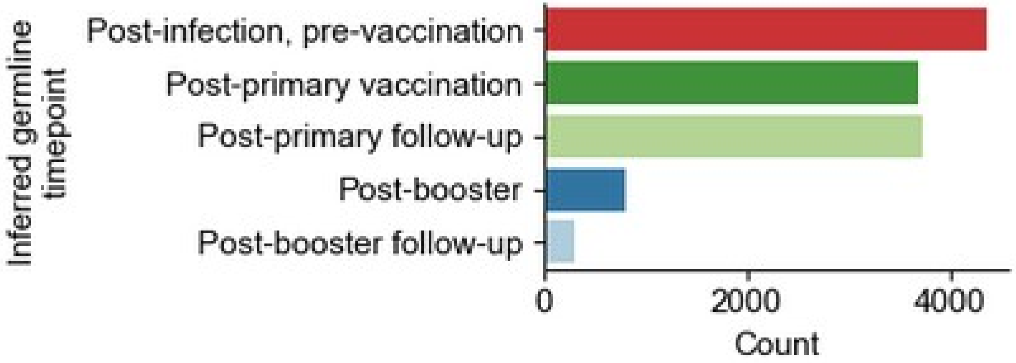


Lineage evolution

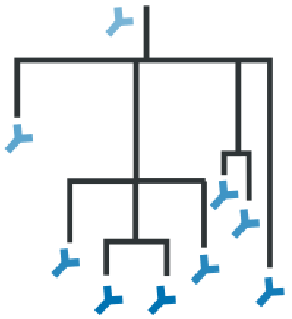




Most clonal lineages appear early

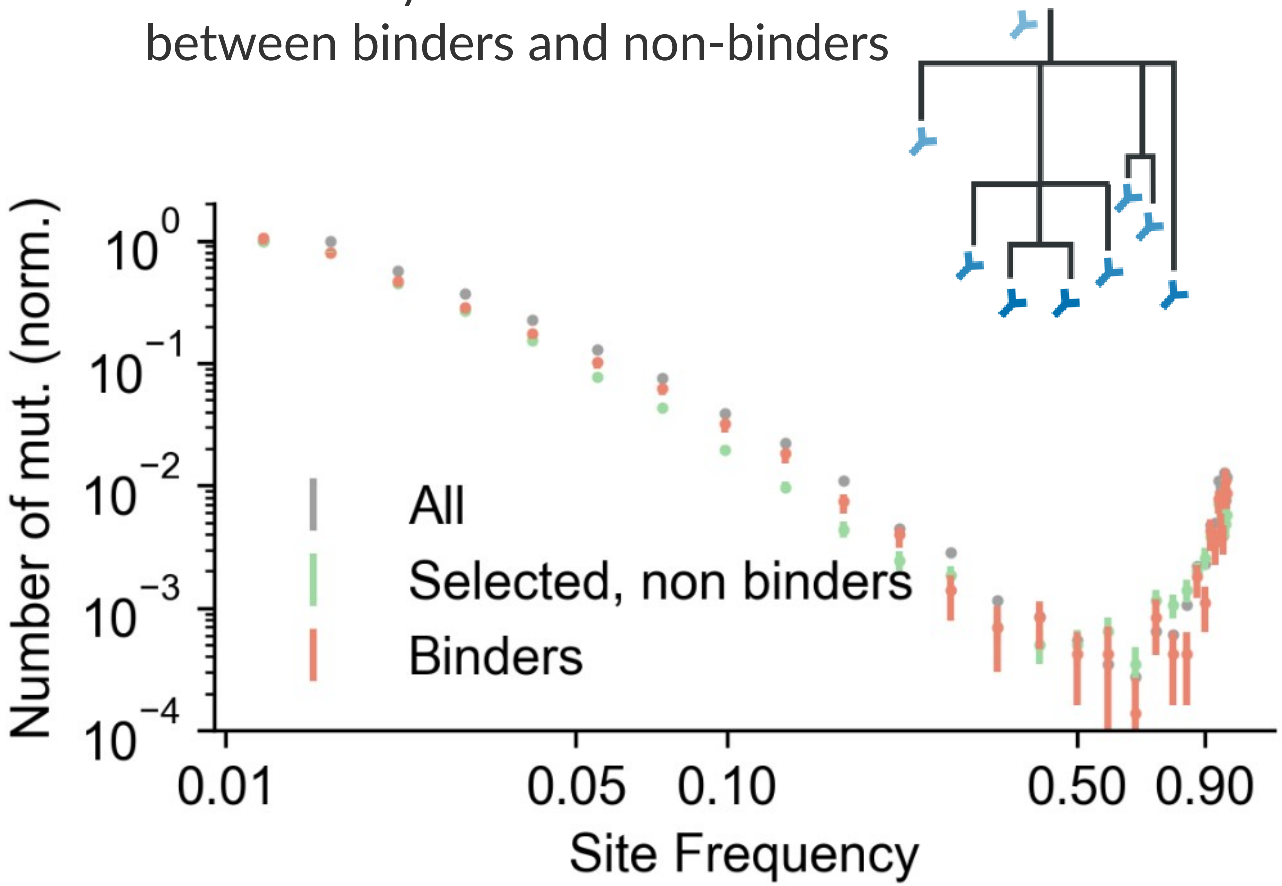


Binder lineages evolve differently



Binder lineages evolve differently

Evolutionary metrics differ
between binders and non-binders



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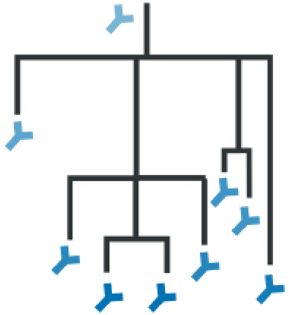

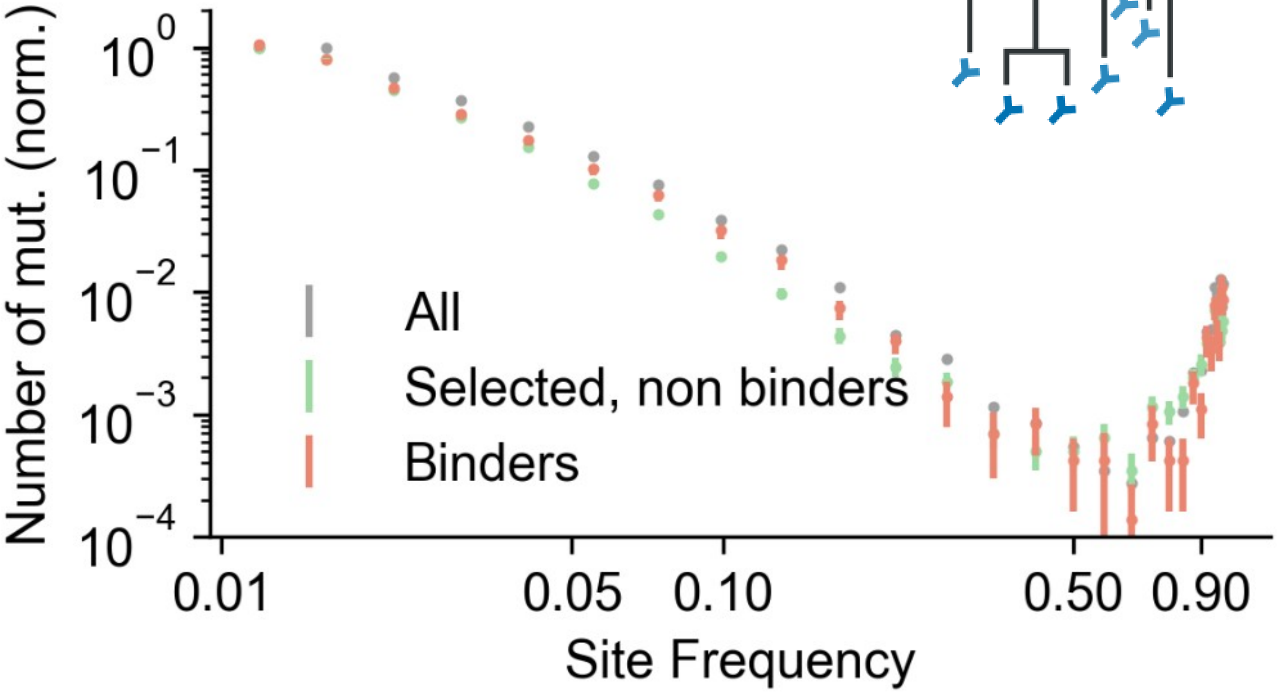

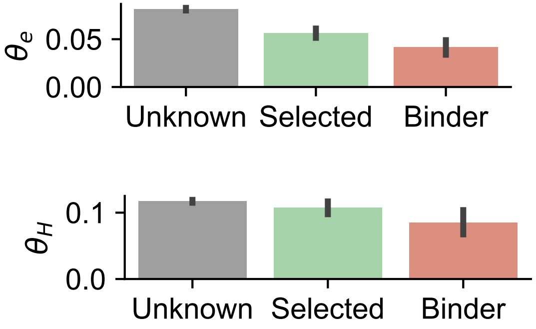

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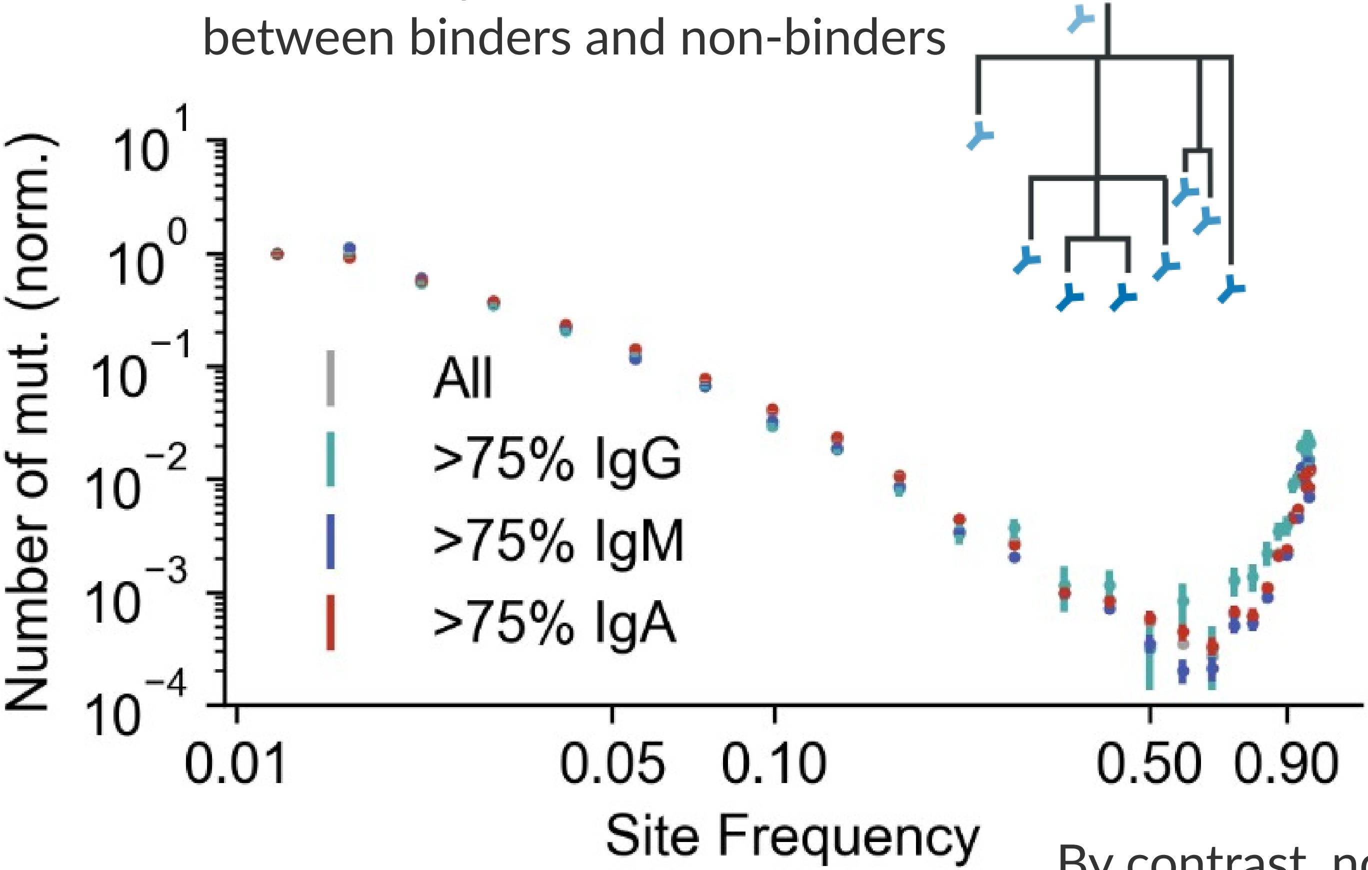
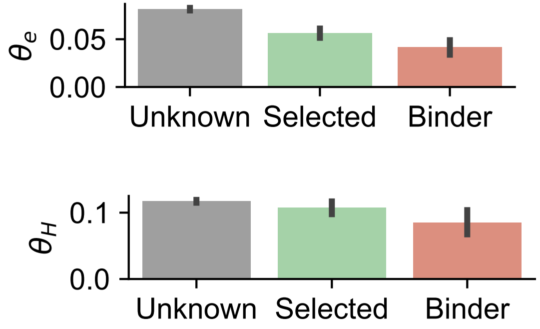


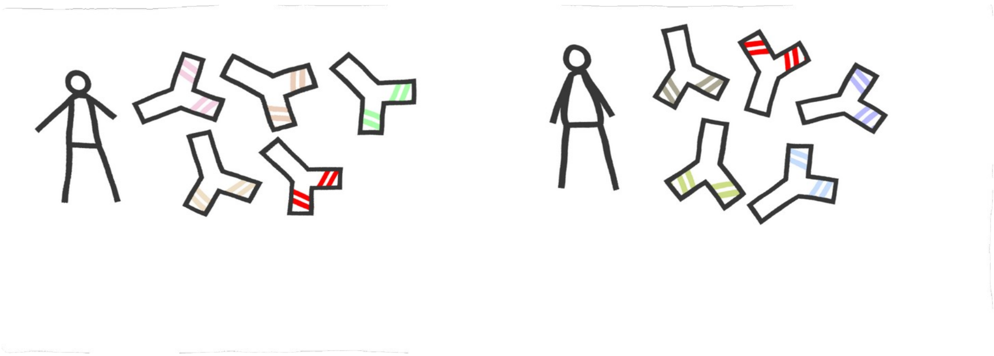


By contrast, no differences
between isotypes

How reproducible is affinity maturation ?

Use identical lineages in different patients as independent replicates

How reproducible is affinity maturation ?



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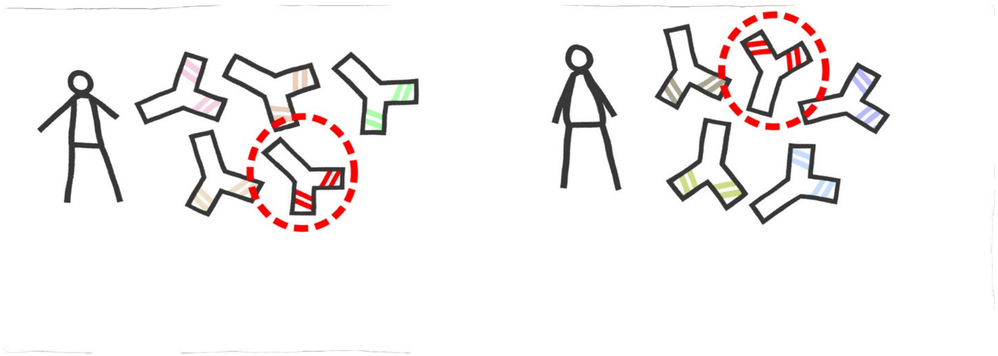

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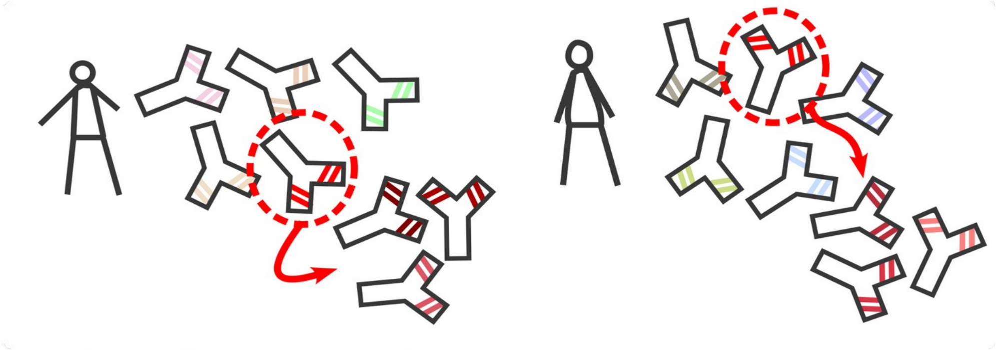

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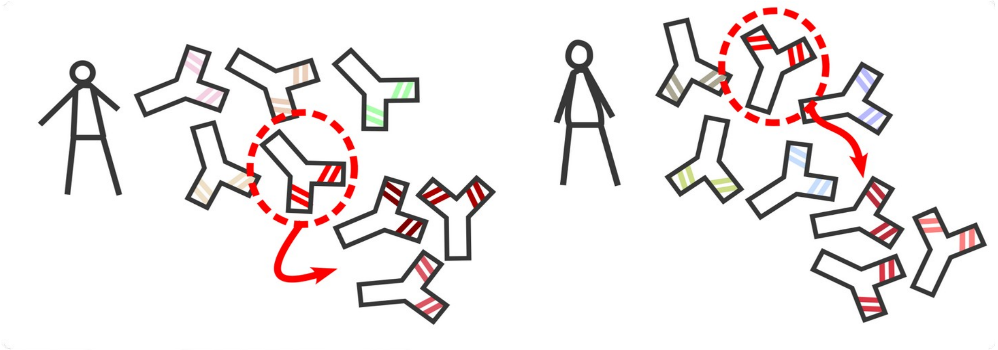
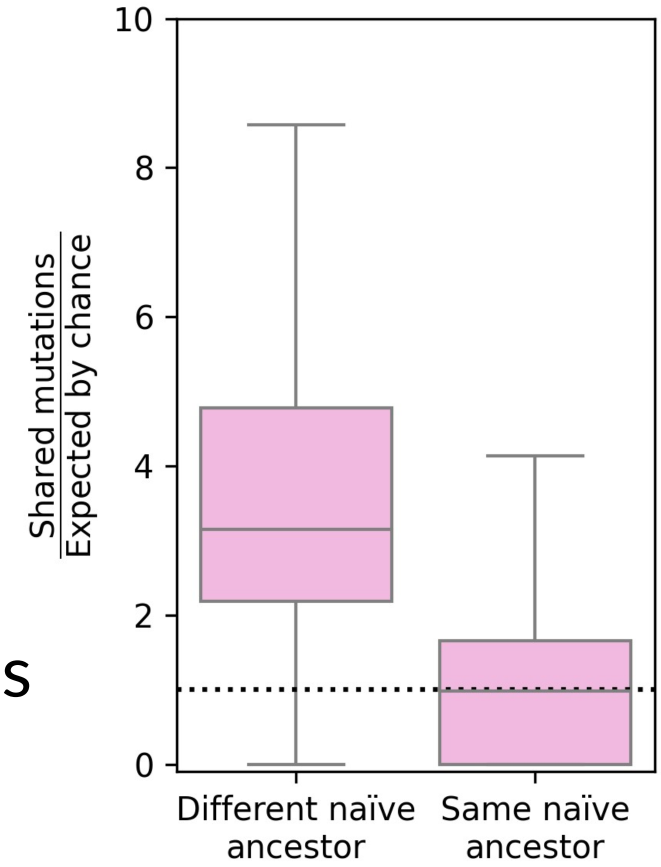


Use identical lineages in different patients as independent replicates

Are B-cells evolving the same way across individuals ?

Mutational Convergence

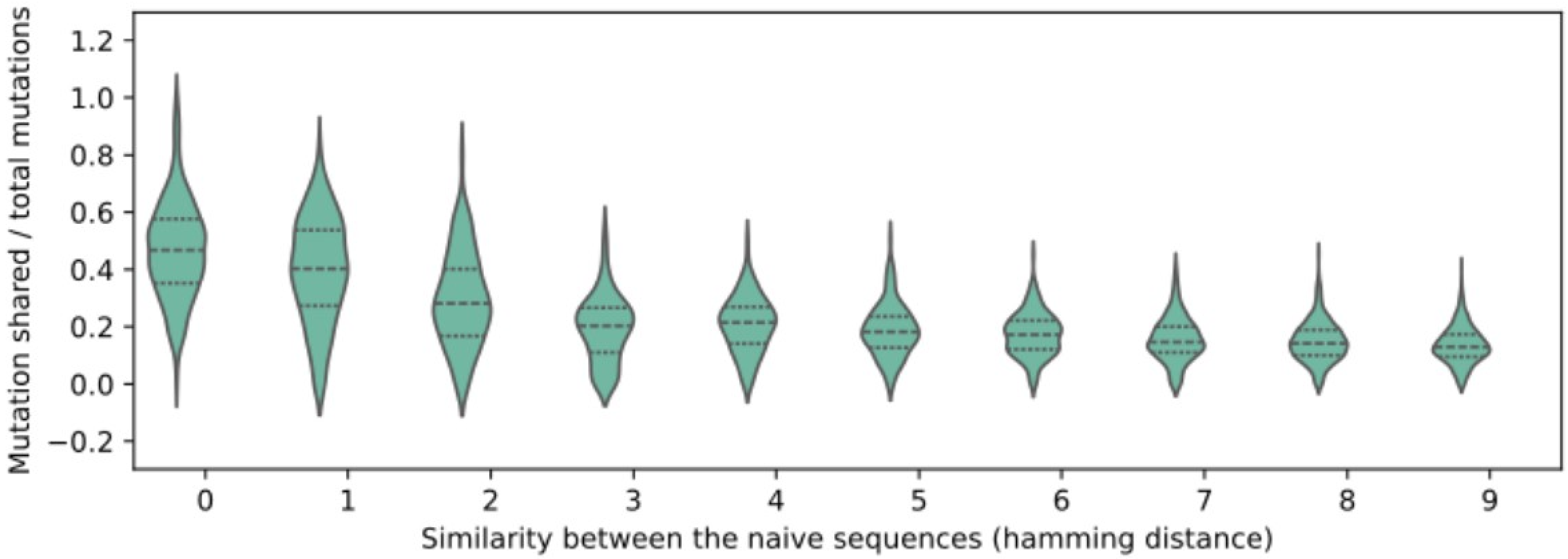
Same V gene



Identical ancestors share more mutations than expected in different individuals

Mutational convergence

What about sequences that are not exactly shared, but similar between individuals ?



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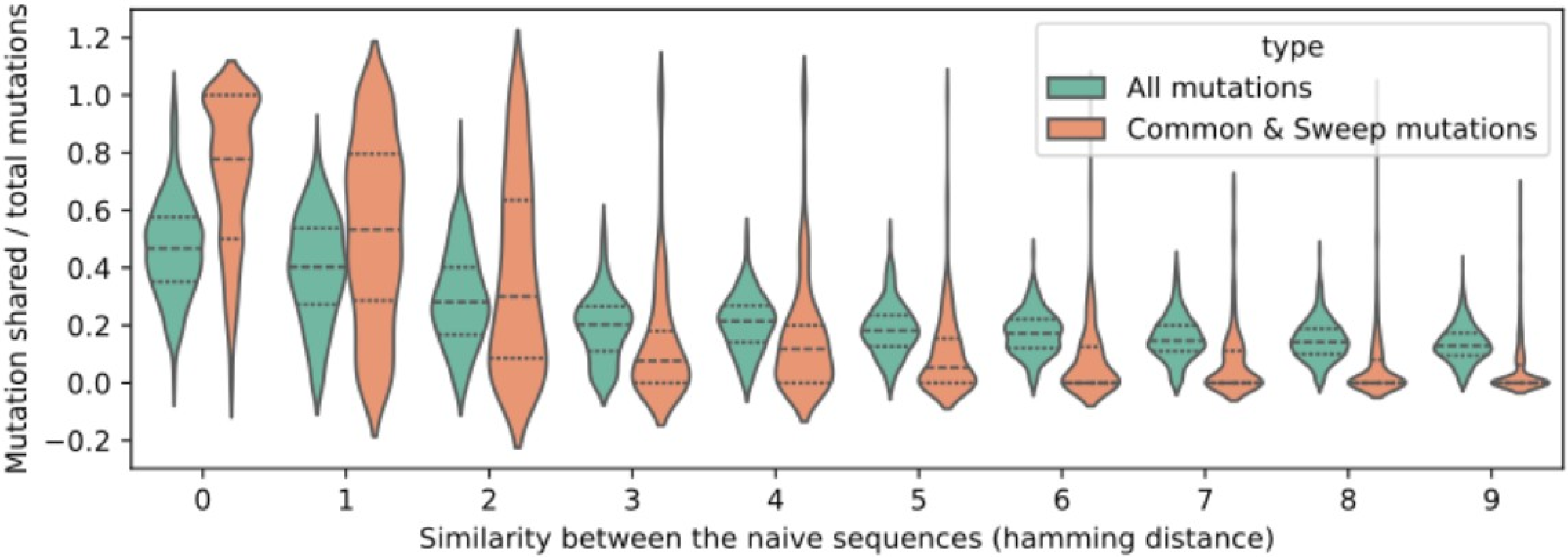

Summary

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- Repertoire are interesting, sequence them !


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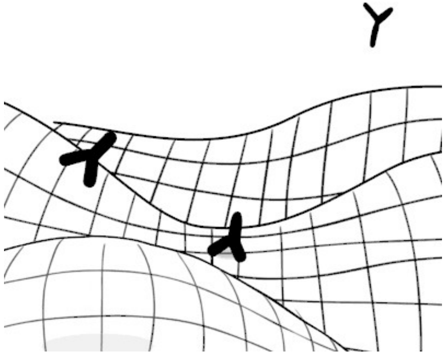

dupic@ciml.univ-mrs.fr